

Reviewed Preprint

v1 • November 25, 2025

Not revised

Reviewed Preprint

v2 • March 3, 2026

Revised by authors

✉ For correspondence:

hugao@njau.edu.cn

† These authors have contributed equally to this work

Competing interests: No competing interests declared

Funding: See [page 23](#)

Reviewing editor: Yuuki Y Watanabe, Graduate University for Advanced Studies, SOKENDAI, Japan

© 2025, Ma et al. This article is distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use and redistribution provided that the original author and source are credited.

Geomagnetic and visual cues guide seasonal migratory orientation in the nocturnal fall armyworm, the world's most invasive insect

Yi-Bo Ma^{1,†}, Gui-Jun Wan^{1,†}, Yi Ji¹, Hui Chen^{1,2}, Bo-Ya Gao¹, Dai-Hong Yu³, Eric J Warrant², Yan Wu⁴, Jason W Chapman^{1,5}, Gao Hu^{1,4} ✉

¹State Key Laboratory of Agricultural and Forestry Biosecurity, College of Plant Protection, Nanjing Agricultural University, Nanjing, China • ²Lund Vision Group, Department of Biology, Lund University, Lund, Sweden • ³Plant Protection Station of Yuanjiang County, Yuanjiang, China • ⁴Key Laboratory of Surveillance and Management of Invasive Alien Species, Guizhou Education Department, Department of Biology and Engineering of Environment, Guiyang University, Guiyang, China • ⁵Centre for Ecology and Conservation, University of Exeter, Penryn, United Kingdom

eLife Assessment

This **fundamental** study presents experimental evidence on how geomagnetic and visual cues are integrated in a nocturnally migrating insect. The evidence supporting the conclusions is **compelling**. The work will be of broad interest to researchers studying animal migration and navigation.

<https://doi.org/10.7554/eLife.109098.2.sa3>

Abstract

The mechanisms guiding nocturnal insect migration remain poorly understood. Although many species are thought to use the geomagnetic field, the sensory basis of magnetic orientation in insects has yet to be clarified. We developed an indoor experimental system to investigate the integration of geomagnetic and visual cues in the seasonal orientation of a globally distributed pest moth, the fall armyworm (*Spodoptera frugiperda*), a highly invasive species which in the past decade has colonized almost all potentially habitable regions of the globe. Our results demonstrate that fall armyworms require both geomagnetic and visual cues for accurate migratory orientation, with visual cues being indispensable for magnetic orientation. When visual and geomagnetic cues are placed in conflict moths become disoriented, although not immediately, indicating that sensory recognition of the conflict requires time to process. We also show that the absence of visual cues leads to a significant loss of flight stability, which likely explains the disruption in orientation. Our findings highlight that visual cues are critical for stable magnetic orientation in the fall armyworm, offering a basis for future investigations of visual-magnetic integration in noctuid migrants.

Introduction

Numerous species of larger nocturnal moths, particularly in the family Noctuidae (hereafter 'noctuid moths'), undertake long-distance multigenerational migrations in the Northern Hemisphere. Every spring across North America and Eurasia, billions of noctuid moths move hundreds of kilometers northwards to summer breeding grounds in the temperate zone, and in the subsequent fall their progeny return to lower latitude wintering areas (Holland et al., 2006 [↗](#); Satterfield et al., 2020 [↗](#); Hu et al., 2016 [↗](#), 2025 [↗](#); Huang et al., 2024 [↗](#)). These migrations take place almost exclusively high above the ground, where fast winds facilitate rapid windborne

transport leading to population redistribution over long distances (Chapman et al., 2010; Hu et al., 2016; Huang et al., 2024). Some of the most abundant species involved in these migrations are the world's most destructive agricultural pests, and thus they are of huge importance for food security and economic prosperity (Bauer & Hoyer, 2014; Satterfield et al., 2020; Guo et al., 2020; Hu et al., 2025). It is thus of paramount importance to understand all aspects of the migratory patterns of noctuid moths.

Tracking the seasonal progression of resources requires noctuid moths to move in the appropriate direction (northwards in spring and southwards in fall), a process entailing three linked steps. Firstly, they determine the seasonally appropriate direction (north or south) in which to travel. Secondly, high-flying noctuid moths select transporting winds that are broadly aligned with this direction. Thirdly, they adopt self-powered flight headings which are more-or-less aligned with the wind but, when required, offset to some degree to correct for crosswind drift (Chapman et al., 2015; Hu et al., 2016). Each step requires use of one or more compass senses based on globally stable cues to determine the required direction and behavioral responses (Mouritsen, 2014).

However, migration at night poses a considerable navigational challenge for noctuid moths and other night-flying insects (Warrant et al., 2011; Mouritsen, 2018; Foster et al., 2018; Gao et al., 2024; Grob et al., 2025). This is because they cannot rely on the sun for compass information in the way that diurnally migrating butterflies and hoverflies do (Mouritsen et al., 2002; Srygley et al., 2008; Gao et al., 2020; Massy et al., 2021; Pakhomov et al., 2025). Despite this challenge, radar, tagging, and tethered-flight studies demonstrate that many larger nocturnal moths are highly sophisticated navigators, capable of selecting and maintaining appropriate movement directions with a high degree of accuracy, even when flying high above the ground (Chapman et al., 2010; Dreyer et al., 2018a,b; Menz et al., 2022; Chen et al., 2023; Dreyer et al., 2025). Their ability to achieve seasonally appropriate migration directions typically surpasses that of diurnal windborne insect migrants (Chapman et al., 2010; Hu et al., 2016; Werber et al., 2025), and even matches the capability of nocturnal songbird migrants (Alerstam et al., 2011; Chapman et al., 2016). Clearly, migratory noctuid moths must possess one or more accurate compass senses, but in all species but one, the source of the compass information has yet to be elucidated (Warrant et al., 2011; Foster et al., 2018; Mouritsen, 2018; Freas et al., 2023; Grob et al., 2025).

The one exception to this lack of knowledge of the sensory basis of navigation in noctuid moths is the Bogong moth (*Agrotis infusa*) of Australia, which uses both a magnetic compass integrated with visual cues (Dreyer et al., 2018a) and a stellar compass (Dreyer et al., 2025) to migrate in seasonally appropriate directions. The Bogong moth is unique, however, among long-range migratory noctuids because a single generation makes bidirectional movements to and from a highly restricted geographic location in southeast Australia (Warrant et al., 2016), rather akin to the migration of eastern North American monarch butterflies to and from a restricted area of central Mexico (Reppert et al., 2016). This is distinct from most other noctuid moth migrants, whose multigenerational migrations involve back-and-forth movements between broad latitudinal zones (Drake & Reynolds, 2012; Chapman et al., 2015; Gao et al., 2020; Tong et al., 2022; Hu et al., 2025) rather than to precise locations. Thus other noctuid moths presumably do not require the same navigational precision, and may therefore be expected to have simpler sensory capabilities than the Bogong moth. We examine this question using the fall armyworm (*Spodoptera frugiperda*), one of the world's most serious crop pests, as a model for understanding general noctuid moth migration and navigation capabilities.

The fall armyworm is a migratory crop pest native to the Americas, where it breeds year-round in tropical regions and annually migrates into temperate regions of North America (Nagoshi & Meagher, 2008; Westbrook et al., 2019). Over the past 10 years, it has invaded and rapidly spread across Africa, Asia, and Australasia (Goergen et al., 2016; Kenis et al., 2023; Tay et al., 2023), assisted by its high migratory capacity (Chen et al., 2022, 2025; Gao et al., 2025), where it causes huge yield losses. Invasive fall armyworm populations now breed year-round in tropical regions of Southeast Asia and South China; each spring they migrate northwards as far as

Northeast China, and the progeny resulting from summer breeding return southwards in the fall (Li et al., 2020 [DOI](#); Sun et al., 2021 [DOI](#); Wu et al., 2021 [DOI](#); Wu et al., 2022 [DOI](#)). We recently found that fall armyworms from the year-round range in Southwest China (Yunnan) exhibit seasonally appropriate migratory headings when flown outdoors in virtual flight simulators, heading northward in the spring and southward in the fall, and this seasonal reversal is controlled by photoperiod (Chen et al., 2023 [DOI](#)). However, the compass mechanism that fall armyworms use to select seasonally appropriate headings remains unknown. The Earth's geomagnetic field is a ubiquitous and reliable source of compass information (Mouritsen, 2018 [DOI](#)), and thus might be expected to be the primary compass cue used by night-flying migratory moths. However, despite intriguing indications of magnetoreception in the oriental armyworm, *Mythimna separata* (Xu et al., 2017 [DOI](#)), so far only the highly specialised Bogong moth has been demonstrated behaviorally to use magnetic information integrated with visual cues for orientation (Dreyer et al., 2018a [DOI](#), Dreyer et al., 2025 [DOI](#)). To test whether more generalist noctuid moth migrants also rely on a magnetic compass for orientation, we investigated an invasive fall armyworm population in China using a newly developed tethered-flight assay designed to quantify migratory orientation (Fig. 1 [DOI](#)). Although highly reductionist relative to natural flight conditions, particularly with regard to the artificial visual cue, this indoor assay provides a controlled framework for isolating the contribution of geomagnetic and visual cues as a first step toward understanding how these signals operate under more realistic ecological scenarios. The results of these experiments are not only of fundamental interest to sensory biology, but are also of applied importance given that the fall armyworm is one of the most mobile and invasive crop pests in the world (Warrant & Dacke, 2011 [DOI](#); Kenis et al., 2023 [DOI](#); Tay et al., 2023 [DOI](#)).

Results

Integration of geomagnetic and visual cues in the seasonal migratory orientation of field-captured fall armyworms

To examine if fall armyworms integrate geomagnetic and visual cues for seasonal migratory orientation, we measured flight responses of tethered moths within a virtual flight simulator (Dreyer et al., 2021 [DOI](#)) using a modified experimental approach used in Bogong moth studies (Dreyer et al., 2018a [DOI](#); Dreyer et al., 2025 [DOI](#)). The simulator consisted of a polyvinyl chloride (PVC) cylinder incorporating a visual cue on the side (a black triangle rising above a black horizon; Fig. 1 [DOI](#)). When tethered within the simulator, the moth is restrained but is free to rotate and thus take up any orientation it chooses. The simulator was placed within a 3D Helmholtz coil system. Under control conditions, the coil was switched off, leaving the Earth's natural magnetic field (NMF) unaltered; under these conditions, magnetic north and geographic north were closely aligned. However, when the coils were switched on, they reversed the horizontal direction of the local magnetic field by 180° (the “Changed Magnetic Field”, CMF) relative to the NMF, while maintaining constant field intensity and inclination angle (Fig. S1 [DOI](#)). Thus the field direction within the experimental arena temporarily switches so that its “local magnetic north” is aligned with geomagnetic south (GMS) and its “local magnetic south” is aligned with geomagnetic north (GMN), but all other field parameters remain constant. The experimental setup consisted of recording moth flight headings across five consecutive 5-minute phases (I, II, III, IV, V) of tethered flight under different experimental conditions that involved changing the azimuthal alignment of the visual cue and the local horizontal magnetic field component with respect to GMN and GMS (Fig. 2 [DOI](#), Fig. S1 [DOI](#)). The assay was divided into these 5-min segments to provide the temporal resolution needed to detect changes in flight orientation as the relative alignment of magnetic and visual cues was systematically altered. In phases I and V moths were exposed to the NMF, whereas in phases II–IV moths were exposed to the CMF (Fig. 2 [DOI](#)). During spring migration trials, the visual cue was aligned with GMN (the expected migratory orientation in spring) at the start of the trial, whereas in fall migration trials the set-up was reversed, with the visual cue initially aligned with GMS (the expected migratory orientation in fall; Fig. 2 [DOI](#)).

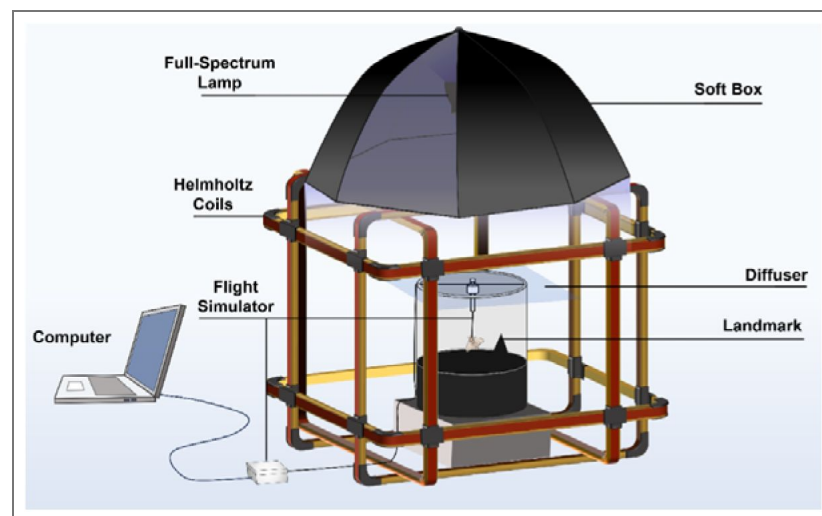


Fig. 1. Schematic of the experimental setup for studying magnetic orientation in fall armyworms.

Moths are tethered to a vertical shaft at the center of the virtual flight simulator, with an encoder recording their flight heading. A full-spectrum lamp illuminates the arena, while the computer controlling the experiment is positioned outside the light field to avoid interference. Moths are free to rotate in any direction during the assay. Full experimental details in Methods. The cylinder is illustrated as clear in the figure to reveal the internal setup, but it is opaque in the actual experiment.

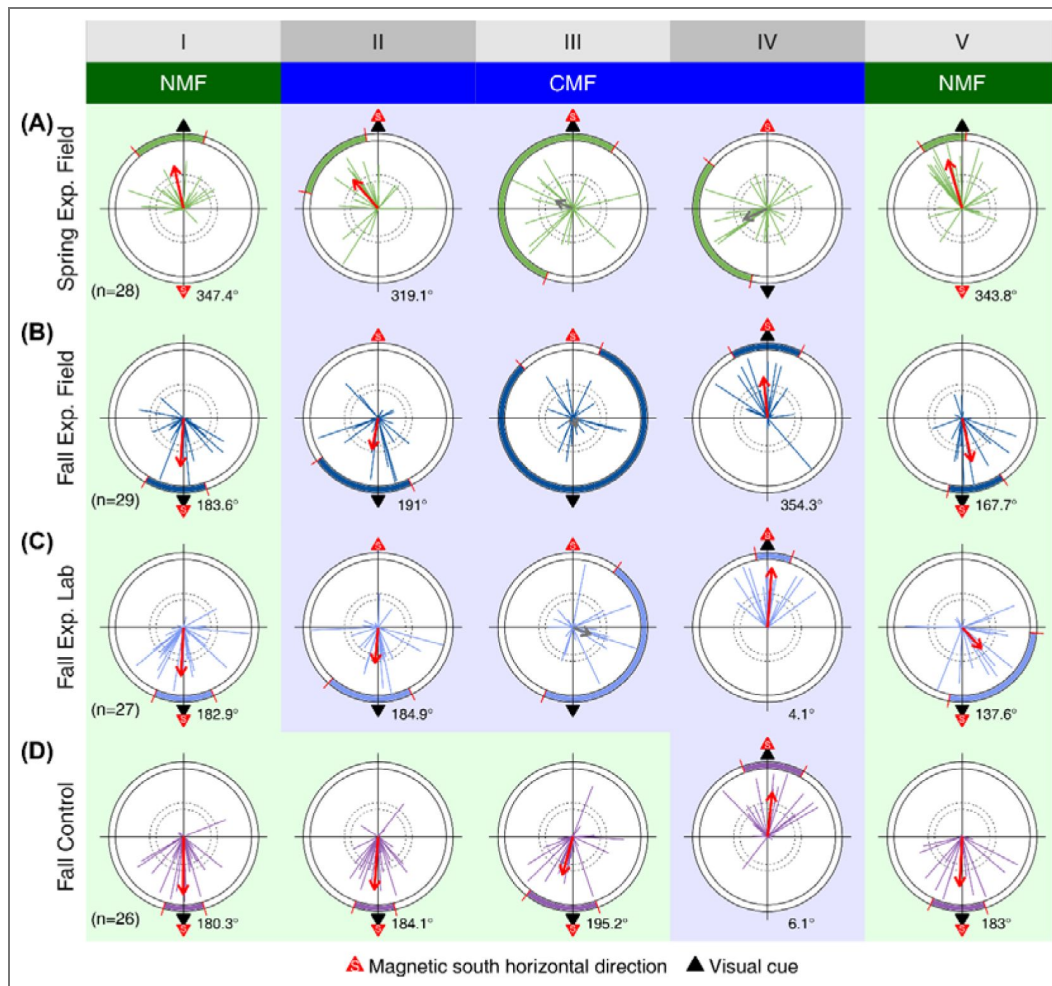


Fig. 2. The Earth's magnetic field and visual cues guide migratory flight behavior in both field-captured and lab-raised fall armyworms.

(A) Flight orientation behavior of spring field-captured moths ("Spring Exp. Field") in response to visual and geomagnetic cues. (B) Flight orientation behavior of fall field-captured moths ("Fall Exp. Field") in response to visual and geomagnetic cues. (C) Flight orientation behavior of lab-raised fall-conditioned moths ("Fall Exp. Lab") in response to visual and geomagnetic cues. (D) Flight orientation behavior of lab-raised control fall-conditioned moths ("Fall Control"), tested with consistent visual and geomagnetic alignment. For simplicity and consistency, we tested only conditions in which the visual cue pointed in the migratory direction, coinciding with magnetic north in spring and with magnetic south in fall. Each group underwent five sequential 5-minute phases (I–V), with each subplot representing individual moths' flight directions in the simulator. The length of each vector represents individual directedness (r), ranging from 0 to 1, where the outer edge of the plot corresponds to $r = 1$. The thick mean vector (MV) arrow represents the weighted average of individual orientations, calculated using Moore's modified Rayleigh test (see Methods), and is red when there is significant group orientation but grey when it is not significant. The R^* value quantifies the directedness of the MV. Dashed circles indicate thresholds for statistical significance, with radii corresponding to $P < 0.05$ and $P < 0.01$. Shaded sections of the outer diameters of the circles represent the 95% confidence limits of the group orientation. The outermost radius represents $R^* = 2.5$. The black triangle denotes the position of the visual cue, while the red triangle indicates the direction of the expected migratory orientation in each season (north in spring, south in fall). The experimental setup included both the natural magnetic field (NMF, panels with light green background) and a changed magnetic field (CMF, panels with light blue background) where the horizontal magnetic field direction was reversed; further details of the magnetic field parameters in the NMF and CMF are shown in Fig. S1.

The first two experiments involved field-captured moths tested during spring and fall migration periods (Figs. 2A and 2B). In phase I, the visual cue was aligned with the expected seasonal magnetic direction in the NMF, and moths exhibited significant group orientation towards the visual cue in both seasons (Spring: mean vector [MV] = 347.4° (95% CI: 319.5°, 16.71°), vector strength [R^*] = 1.59, $P < 0.001$, Fig. 2A-I; Fall: MV = 183.6° (95% CI: 211°, 162°), R^* = 1.76, $P < 0.001$; Fig. 2B-I). In phase II, the horizontal component of the geomagnetic field was rotated 180° (the CMF condition), creating a conflict between the visual cue direction and the expected magnetic orientation. Despite the shift in the magnetic field direction, moths continued to show significant group orientation toward the visual cue during this 5-minute period (Spring: MV = 319.1° (95% CI: 351°, 283°), R^* = 1.39, $P < 0.005$, Fig. 2A-II; Fall: MV = 191.0° (95% CI: 235°, 153°), R^* = 1.16, $P < 0.05$, Fig. 2B-II), indicating that, at least initially, the visual cue was dominant compared to the magnetic cue. During phase III, which was a second 5-minute period of the experimental conditions applied in phase II, moths lost their significant group-level orientation in both seasons (Spring: R^* = 0.67, $P > 0.05$, Fig. 2A-III; Fall: R^* = 0.34, $P > 0.05$; Fig. 2B-III), indicating that over time they had become confused due to the conflicting nature of the cues. Although we did not analyze flight activity level due to technical limitations, the loss of group orientation is unlikely to be attributable to a decline in the moths' motivation to maintain orientation during prolonged flight in the simplified experimental environment, as the Rayleigh test showed no significant differences in the r -values of individual flight vectors across phases I–III (Figs. 2A and 2B; Table S2; Table S3). In phase IV, the visual cue was realigned with the expected seasonal magnetic orientation, but this time in the CMF (i.e., the cues were arranged in the same way as in phase I but rotated by 180°). Therefore, the moths should not be able to distinguish between phase I and phase IV and thus are expected to show the same orientation response in both phases. Indeed, the moths tended to show group orientation towards the congruent cues once again, albeit not quite reaching significance in spring (Spring: MV = 245.2° (95% CI: 193.2°, 309.4°), R^* = 0.95, $0.05 < P < 0.10$, Fig. 2A-IV; Fall: MV = 354.3° (95% CI: 331.4°, 27.2°), R^* = 1.49, $P < 0.005$; Fig. 2B-IV). Finally, in phase V, the initial configuration was restored, and moths regained significant group orientation toward the congruent visual cue and the expected magnetic orientation in the NMF (Spring: MV = 343.8° (95% CI: 326.56°, 2.92°), R^* = 1.84, $P < 0.001$, Fig. 2A-V; Fall: MV = 167.7° (95% CI: 191°, 146°), R^* = 1.62, $P < 0.001$, Fig. 2B-V). These results demonstrate that fall armyworm integrates geomagnetic field information with visual cues to achieve stable orientation. However, when geomagnetic and visual cues do not align with expected seasonal directions, moths gradually lose orientation, reinforcing the critical role of cue integration in maintaining migratory stability.

Integration of Geomagnetic and Visual Cues Is Consistent in Laboratory-Reared Fall Armyworms

Our previous research showed that fall armyworms reared under artificially simulated fall conditions and tested indoors with optic flow and starry sky projections exhibited southward orientation in a flight simulator, consistent with the behavior observed in wild individuals tested outdoors under natural night skies (Chen et al., 2023). To determine whether lab-raised moths exposed to simulated seasonal photoperiods respond the same way to geomagnetic and visual cues, we tested a population reared under a simulated fall photoperiod in the same experimental setup as that of the field-captured fall group (Fig. 2C). The results closely matched those from the field-captured wild population (Fig. 2B). In phase I, with visual cues aligned to GMS in the NMF, lab-raised moths oriented significantly southward (MV = 182.9° (95% CI: 203°, 156°), R^* = 1.77, $P < 0.001$, Fig. 2C-I). When the geomagnetic field was reversed (CMF), moths initially orientated towards the visual cue, showing no immediate response to the shifted field (MV = 184.9° (95% CI: 222°, 153°), R^* = 1.31, $P < 0.05$, Fig. 2C-II). After an additional 5-minute period without changes, significant group-level orientation was lost (R^* = 0.69, $P > 0.05$, Fig. 2C-III). When the visual cue was realigned with the changed GMS, significant orientation was restored (MV = 4.1° (95% CI: 350.7°, 19.3°), R^* = 2.17, $P < 0.001$, Fig. 2C-IV), though the direction was opposite that in phase I. Finally, returning to the original setup restored the initial orientation pattern (MV = 137.6° (95% CI: 189.9°, 94.9°), R^* = 1.04, $P < 0.05$, Fig. 2C-V). These results demonstrate that lab-raised

fall armyworms integrate geomagnetic and visual cues similarly to wild populations, emphasizing the role of photoperiod during development in shaping seasonal multimodal migratory orientation.

To confirm that the loss of orientation in experimental groups was due to the cue conflict rather than diminished motivation to maintain orientation arising from a prolonged test duration, we conducted a control experiment with the lab-raised simulated fall population, where the visual cue and GMS remained consistently aligned. In this group, moths maintained significant orientation in their expected migratory direction throughout the experiment when exposed to congruently-aligned cues, irrespective of whether under NMF or CMF conditions (Fig. 2D; directional statistics in Table S1). This confirms that the loss of orientation observed in experimental groups was driven by misalignment between visual and geomagnetic cues.

Fall armyworms show a delayed response to changes in magnetic fields

Our results show that migratory fall armyworms rely on both the Earth's magnetic field and visual cues to determine their flight direction. However, for moths to maintain constant orientation, the two sensory inputs must remain congruent, and when they come into conflict the moths become disoriented at the group level (but not at the individual level; Fig. 2). Notably, this response to the conflict between visual and geomagnetic cues was not instantaneous but delayed until the second 5-minute experimental phase with this condition, consistent with similar results from Bogong moths (Dreyer et al., 2018a). Initial analyses using a 5-minute resolution provided limited temporal detail. To improve resolution, we subdivided each 5-minute phase into ten 30-second intervals and independently analyzed directional consistency using Moore's Rayleigh test to calculate the R^* for each 30-second period (Fig. 3).

During phase I, R^* values in both experimental groups remained >1 (Figs. 3A-I and 3B-I), indicating significant orientation behavior. When the magnetic field was rotated to conflict with visual cue directions, R^* values did not drop immediately but instead declined gradually over time (Figs. 3A-II and 3B-II). By phase III, R^* values were consistently <1 , indicating a loss of significant orientation (Figs. 3A-III and 3B-III). Upon realigning the magnetic field with visual cues, R^* values showed a gradual increase, eventually exceeding 1 in both groups (Figs. 3A-IV and 3B-IV). When the original configuration was restored, the field-captured group consistently exhibited R^* values above 1 (Fig. 3A-V), while the lab-raised group had R^* values hovering around 1 (Fig. 3B-V). These results indicate that fall armyworms require time to resolve conflicting navigational inputs before losing group orientation entirely, or to react to congruent cues and regain group orientation. In the lab-raised control group, where the geomagnetic field and visual cues remained consistently aligned, R^* values consistently remained above 1 (Fig. 3C), confirming that the decline in group-level orientation observed in the experimental groups was driven by conflicts between geomagnetic and visual cues, mirroring the results found in Bogong moths (Dreyer et al., 2018a).

Visual cues are essential for magnetic orientation in fall armyworms

Growing evidence supports the idea that geomagnetic cues provide essential compass information for orientation across a broad range of taxa, supporting diverse navigational tasks (Mouritsen, 2018; Dreyer et al., 2018a; Fleischmann et al., 2018; Wan et al., 2021; Grob et al., 2024; Goforth et al., 2025). To determine whether geomagnetic input alone is sufficient for flight orientation in fall armyworms, we conducted a series of experiments to evaluate the role of visual cues in their orientation behavior. In particular, we examined whether these moths could maintain orientation in the absence of visual cues.

As a control, we first tested lab-raised, fall-conditioned moths in an arena where visual landmarks were aligned with GMS in the NMF (Fig. 4A). When illuminated, moths showed significant group orientation towards the visual cue ($MV = 218.48^\circ$ (95% CI: 242° , 187°), $R^* = 1.50$, $P < 0.005$, Fig. 4A-

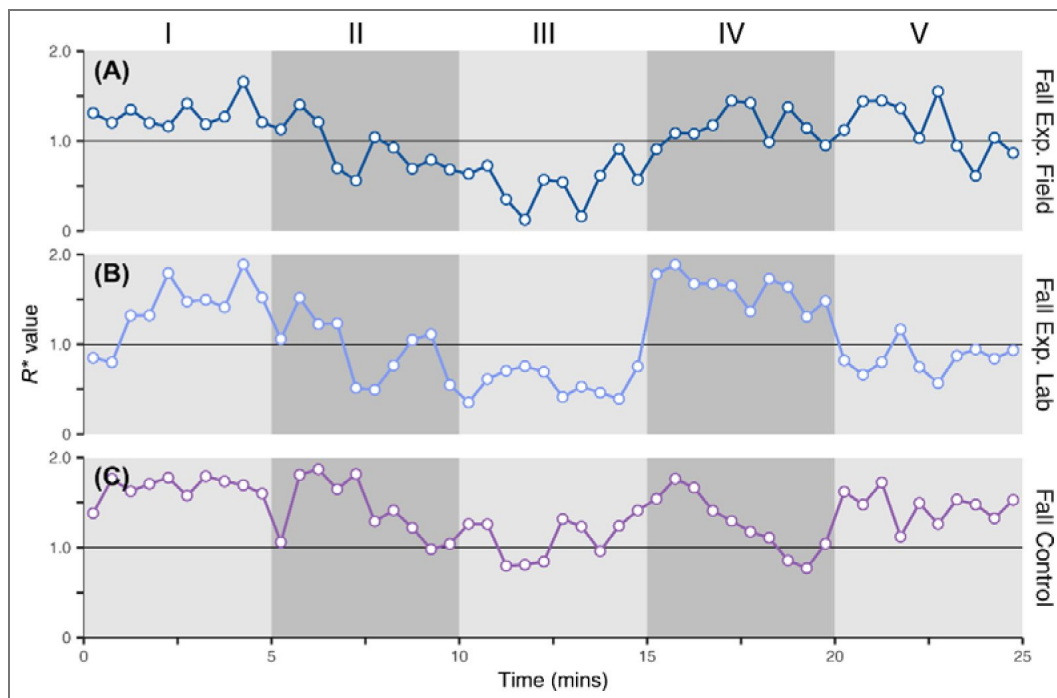


Fig. 3. Fall armyworms exhibit a delayed response to changes in magnetic-visual cue alignment.

The behavioral data for (A) fall field-captured experimental ("Fall Exp. Field", data from Fig. 2B), (B) fall lab-raised experimental ("Fall Exp. Lab", data from Fig. 2C), and (C) fall lab-raised control ("Fall Control", data from Fig. 2D) groups were analyzed in 30-second time bins, resulting in ten bins over each 5-minute phase of the experiment. For each group, Moore's modified Rayleigh test was applied, and the obtained R^* values were plotted against time. $R^* > 1$ indicates a significant collective orientation within that 30-second interval, while $R^* < 1$ indicates the absence of significant group-level orientation.

I²), consistent with our previous results (Fig. 2B). To determine whether moths could maintain orientation without visual information, we then removed all light sources, allowing moths to continue flying in total darkness. Under these conditions, moths lost significant group-level orientation ($R^* = 0.5$, $P > 0.05$, Fig. 4A-II), suggesting that their orientation behavior is dependent on the presence of light. The loss of orientation may be due to the absence of usable visual landmarks, or the lack of light in a dark environment causing the moths to lose their sense of direction. However, it is still unclear how the absence of usable visual landmarks in non-dark environments would affect the moths' orientation behavior.

To further investigate this, we tested a lab-raised, fall-conditioned population of moths in a simulator, primarily to determine whether moths could maintain orientation under illuminated conditions where visual cues on the interior walls of the cylindrical flight arena were minimized. In this experiment, we provided a uniform visual environment (i.e., under illuminated conditions, visual cues were not provided on the interior walls of the cylindrical flight arena). After recording 10 minutes of orientation behavior under the NMF, we reversed the horizontal component of the geomagnetic field using Helmholtz coils (CMF) and recorded their flight orientation for an additional 10 minutes. Despite the altered magnetic field direction and sufficient time for adaptation, moths exhibited no significant group orientation across all four testing phases (Fig. 4B; directional statistics in Table S1). The lack of group-level orientation under near-uniform visual conditions suggests two possibilities: individual moths may have maintained stable flight, but due to the absence of reliable orientation cues, their group-level directional choices were random; or their flight may have been unstable, making it difficult to assess individual orientation behavior. We first compared the differences in the Rayleigh test r -values of moths across different experimental phases (Fig. 4C, Left), using the Wilcoxon signed-rank test for the non-independent Fall Exp and Fall Dark groups, and the Wilcoxon rank-sum test for all other independent group comparisons. This is because the Rayleigh test evaluates the concentration of data, with an r -value close to 1 indicating highly consistent flight directions, and close to 0 suggesting more random flight directions. The results showed that under conditions with visual cues (Fall Exp. Dark-I), moths had significantly higher r -values compared to conditions of complete darkness (Fall Exp. Dark-II) or visual cues reduced to the bare minimum (Fall Exp. BMVC).

We also considered the limitations of the Rayleigh test, as its r -value only reflects the overall directional tendency and does not reveal real-time flight stability. For example, a moth exhibiting two stable flight phases, one oriented north and the other south, would yield a low r -value despite maintaining stability within each phase. Conversely, a moth losing flight stability could also produce a low r -value, making it difficult to distinguish between these two scenarios based on r -value alone. Therefore, we introduced the analysis of per-second angular change (Fig. 4C, Right) to assess dynamic fluctuations in flight behavior, providing a clearer understanding of whether the observed differences in r -values were due to stable orientation or a loss of flight control. Our analysis showed that the experimental groups tested under conditions lacking visual cues exhibited significantly lower flight stability than the other experimental groups (Fig. 4C; Fig. S4). Moths in these conditions exhibited significantly higher angular change rates, indicating a greater tendency to alter flight direction and an inability to maintain stable orientation. Notably, even though a small subset of individuals in these groups managed to sustain stable flight, their directional choices appeared random rather than goal-oriented (Figs. 4A-II and 4B).

Discussion

The integration of geomagnetic and visual cues for flight orientation has so far been experimentally demonstrated only in the Australian Bogong moth, a species undertaking a precisely oriented, long-distance round-trip migration to and from a highly restricted aestivation site within a single generation (Dreyer et al., 2018a). Although this life-history strategy is uncommon among migratory insects, our development of an indoor behavioral paradigm in the fall armyworm, a globally distributed nocturnal migratory insect with multi-generational and partially migratory patterns in China, extends this dual-cue orientation strategy to a species exhibiting a migratory ecology more typical of insects in general. These findings demonstrate that

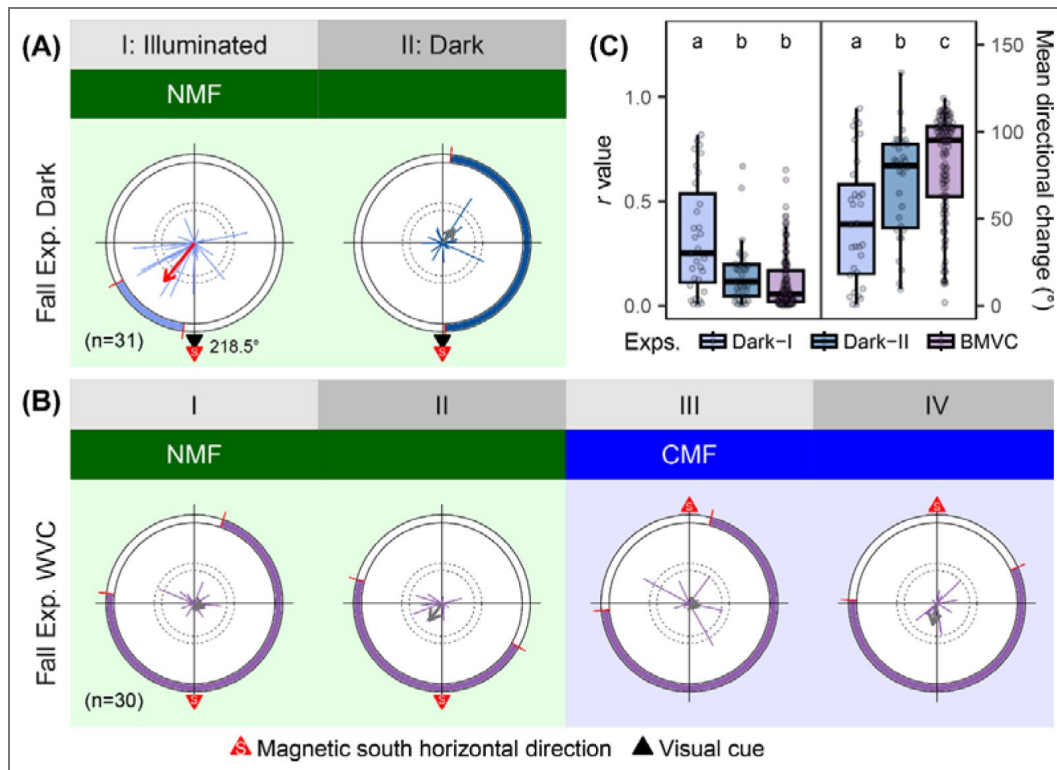


Fig. 4. Visual information is essential for maintaining group flight orientation in fall armyworms.

(A) The lab-raised, fall-conditioned population lost significant group orientation under complete darkness ("Fall Exp. Dark"). (B) The lab-raised, fall-conditioned population exhibited a significant loss of group orientation under illuminated conditions where visual cues were reduced to the bare minimum ("Fall Exp. BMVC"), i.e., obvious visual cues such as the black triangle shown in Fig. 1 were not provided; however, because the experiments were conducted under illuminated conditions, complete elimination of all visual information is impossible. (C) The distribution of Rayleigh test *r*-values for individual moth orientations across different treatment groups (Left), and the distribution of average directional change per second (Right), the latter reflecting flight stability. The box plots represent the interquartile range (IQR), with the horizontal line inside indicating the median. Whiskers extend to the most extreme data points within 1.5 times the IQR. Pairwise comparisons of the *r*-values were performed using the Wilcoxon rank-sum test and Wilcoxon signed-rank test (The Wilcoxon signed-rank test is only used to compare non-independent (i.e., paired or related) samples. For more information, please refer to the Statistical Analysis section.), with multiple comparisons corrected by the Benjamini-Hochberg method. Detailed statistical results are provided in Table S4. Comparisons of the average directional change per second were performed using same method as that used for *r* value, with statistical results provided in Table S5. Groups labeled with different letters differ significantly ($P < 0.05$). We also analyzed the *r*-values and average directional change per second in the Fall Exp. Field, Fall Exp. Lab, Exp. Fall Control, and the experiment shown in this figure, with results consistent with those shown here (see Tables S4 and S5, Fig. S4).

this orientation mechanism is not unique to the Bogong moth and may instead represent a conserved orientation mechanism broadly employed across migratory moth species, though additional species will need to be examined. Crucially, we show that visual cues are indispensable for magnetic orientation in the fall armyworm, as directional responses disappeared in the absence of structured visual input. Furthermore, our findings reinforce recent results on seasonal shifts in orientation behavior in this species (Chen et al., 2023 [DOI](#)), highlighting the ecological relevance of this guidance system.

Our findings thus emphasize the importance of integrating multiple cues for successful orientation. It is often assumed that animals lack magnetic orientation capabilities if they fail to orient under changes in the geomagnetic field (Pakhomov et al., 2025 [DOI](#)), either in appropriate lighting conditions or in darkness, without additional cues. However, our study demonstrates that the absence of group seasonal orientation under geomagnetic fields, in full-spectrum light or in darkness, does not preclude magnetic orientation when additional sensory cues are integrated. Specifically in the Northern Hemisphere, a magnetic compass provides global directional cues for fall armyworms, enabling them to align magnetic north (or magnetic pole) and south (or magnetic equator) with visual cues during spring and fall, respectively, to facilitate seasonal orientation—a hallmark of annual long-distance migration in migratory species. Although the magnetic compass is critical, as evidenced by the loss of group orientation when the horizontal geomagnetic field was reversed while intensity and inclination remained unchanged, our results reveal that magnetic orientation cannot occur without appropriate visual cues. We also show that the loss of flight stability observed when visual cues are minimal is likely a major cause of the lack of group magnetic orientation. This highlights the complexity of magnetic orientation strategies in nocturnal migratory insects, emphasizing the crucial interaction between the magnetic compass and visual navigation systems. Further investigation by altering the geomagnetic polarity and vertical geomagnetic component is required to determine whether the magnetic compass guiding seasonal orientation in nocturnal migratory fall armyworms is polarity-sensitive, inclination-sensitive or both, which is pivotal to understanding the biophysical mechanisms underlying magnetoreception (Mouritsen, 2018 [DOI](#); Hore et al., 2016 [DOI](#); Grob et al., 2024 [DOI](#)).

Desert ants and Bogong moths have also been shown to integrate geomagnetic and visual cues for navigation. Desert ants rely on a magnetic compass during look-back-to-the-nest behavior, combining magnetic cues with local visual landmarks, such as their nest (Grob et al., 2024 [DOI](#)). In contrast, Bogong moths have been demonstrated to integrate global stellar cues with the Earth's magnetic field for orientation (Dreyer et al., 2025 [DOI](#)). Like Bogong moths, fall armyworms can migrate hundreds, or even thousands, of kilometers over several successive nights (Warrant et al., 2016 [DOI](#); Li et al., 2020 [DOI](#)). In nature, they are likely to encounter a range of luminance-gradient visual cues, including relatively stable celestial cues as well as transient or shifting local features encountered en route. Although such natural cues differ from our simplified laboratory stimulus, they may represent intermittently sampled visual inputs that can be optimally integrated with magnetic information, with the congruency between visual and magnetic cues likely playing a key role in maintaining a stable compass response. Whether the cues are static or changing, brief periods without them may still allow the subsequent recovery of a stable long-distance orientation strategy. Determining which types of natural visual cues support the magnetic-visual compass, and how they interact with magnetic information, including how their momentary alignment or angular relationship is integrated and how such visual cue–magnetic field interactions may require time to influence orientation, together with elucidating the genetic and ecological bases of multimodal orientation, will be important objectives for future research.

In conclusion, our findings underscore the indispensability of visual cues in facilitating seasonal magnetic orientation and, unlike some vertebrates that can rely solely on magnetic information (Mouritsen, 2018 [DOI](#)), highlighting the complex, multimodal nature of navigation in nocturnal migratory insects. This work advances our understanding of migratory orientation strategies and raises the possibility that similar magnetic-visual integration mechanisms may be shared among long-distance migratory moths. Furthermore, the application of genome editing in fall

armyworms, combined with the indoor behavioral paradigm developed here, provides a promising avenue for dissecting the molecular basis of magnetoreception and the interplay between sensory systems.

Methods

Field population

All wild populations used in this study were collected from farmland in Yuanjiang Hani and Yi Autonomous County, Yunnan Province, a region primarily characterized by maize cultivation. Fall-generation populations were collected during September to October of 2023 and 2024, while spring-generation populations were collected from April to May in 2024, as mature larvae at the 5th to 6th instar stages. Each larva was individually collected and reared in a cylindrical container, fed with fresh maize leaves from the collection site until pupation. Upon pupation, larvae were transferred to transparent, thin plastic cups sealed with plastic film. A cotton ball moistened with water was placed in each cup to maintain adequate humidity during the pupal stage. After eclosion, adults were individually provided with a 10% honey solution. All experimental individuals were reared at the Yuanjiang Plant Protection Station, a facility located near agricultural fields and well isolated from urban light pollution. To maintain natural environmental conditions, all electrical devices were strictly excluded from the facility, and windows were kept open around the clock to allow for natural light cycles and ambient temperature conditions.

Lab-raised population

The laboratory-reared population of the fall armyworm used in this study was primarily derived from wild individuals collected from a maize field in Yuanjiang, Yunnan Province (Google Maps coordinates: 23.604°N, 101.977°E) during October to November 2022, with the exception of individuals used in experiments Fall Exp. Dark and Fall Exp. BMVC (see Fig. 4 [↗](#)), which were collected from the same field location during October to November 2024. All experiments were conducted using individuals from the 3rd to 5th laboratory-reared generations.

For behavioral assays conducted under ambient temperature conditions, a programmed photoperiodic regimen was implemented to simulate the seasonal changes in day length typical of autumn in East Asia. All treatments were maintained at a constant temperature of $27 \pm 1^\circ\text{C}$ and a relative humidity of $60\% \pm 5\%$. In the fall photoperiod treatment, the initial light cycle at egg hatching was set to 13 hours of light and 11 hours of darkness (13L:11D), followed by a daily reduction of 2 minutes in the light period. By the time adult moths emerged and were subjected to experimental testing approximately one month later, the photoperiod had been adjusted to approximately 12L:12D.

Attachment of tethering stalks on moths

We used the same method for the attachment of tethering stalks on moths as in the previous study on fall armyworm moth orientation (Chen et al. 2023 [↗](#)). Prior to tethering, unmated 2-day-old adult moths were housed in plastic cups and sedated at 4°C for at least 1 minute to facilitate handling. The moths were then transferred to an operational platform, where scales were carefully removed from the junction of the dorsal thorax and abdomen to ensure secure attachment. The tether consisted of a slender, non-magnetic copper stalk measuring 0.75 mm in diameter and approximately 2 cm in length. The terminal end of the stalk was bent into a fork-like structure, which was carefully affixed to the prepared thoracic-abdominal junction using Pattex PSK12CT-2 glue. To maintain consistency, this standardized tethering method was applied uniformly across all experimental protocols.

Behavioral apparatus

Our flight simulator system (Dreyer et al., 2021 [↗](#)) is consistent with the one used in previous research on fall armyworm (Chen et al., 2023 [↗](#)), utilizing the Flash flight simulator system; however, in contrast to previous studies, we did not include optic flow in order to avoid introducing additional visual complexity. The Flash flight simulator system was developed based on the early design of the Mouritsen-Frost flight simulator, and adapted for our experiments conducted in Yuanjiang (longitude 101.98°E, latitude 23.60°N). The moths are able to freely rotate on the horizontal plane within the simulator, with their azimuth measured by an encoder system with a resolution of 0.9°. The flight direction (relative to magnetic north) is recorded in real time by the Flash flight simulator data acquisition system developed by Hui Chen, and the data is saved as angle values in a text file. In our experiments, the encoder (made of non-magnetic materials) samples the azimuth five times per second and is equipped with a graphical interface, which allows for continuous monitoring of the moth's azimuth and flight status to reconstruct the flight path. The encoder is connected to the axis that holds the moth, with the axis measuring 15 cm in length and 1 mm in diameter, made of carbon fiber material and wrapped in a carbon fiber tube to protect its structure.

The arena consists of an opaque PVC cylinder with a diameter of 400 mm, a height of 500 mm, and a thickness of 5 mm as the main structure, surrounded by white wallpaper, with a 20 cm high black wallpaper strip at the bottom simulating the horizon. A visual cue is provided by a black isosceles triangle (10 cm high, 10 cm base) made from black wallpaper and fixed to the horizon at the bottom of the arena. The entire arena is placed on a square wooden board with a side length of 50 cm and a thickness of 1 cm, covered with black blackout cloth. The bottom of the wooden board is equipped with a plastic base to reduce friction, ensuring that the arena can rotate smoothly and quickly during the experiment, changing the direction of the visual cues while avoiding interference with the moth's activity.

In the BMVC experiment, we used an opaque white acrylic cylinder with the same dimensions as the PVC cylinder as the arena. No modifications were made to the interior of the arena, primarily because the interior of the white acrylic cylinder is smooth and uniform enough, and any additional treatment could potentially introduce noticeable visual cues. We also removed the carbon fiber crossbar used to fix the simulator and instead fixed it directly to the top acrylic cover to further minimize potential visual cues that the moth might use. Our goal was to minimize the visual cues available to the moth as much as possible, rather than attempting to completely eliminate all visual cues.

A DH1766-series linear, three-channel, programmable DC power supply (Beijing Dahua Radio Instrument Factory, China) with adjustable current and voltage was used to power the full-spectrum LED employed in the experiments. The light source was enclosed within a 90 cm diameter GODOX softbox to ensure even distribution of light within the experimental arena. The softbox is fixed to a Helmholtz coil, with the flight simulator located at its center. The top of the arena is equipped with a UV-transmitting acrylic panel and covered with Lee Filters 250 semi-white diffusion paper to control the amount of light transmitted. Spectral measurements, which include both spectral irradiance and composition, were taken at the position corresponding to the moth's head using an ATP2000P spectrometer (Optosky Technology Co. Ltd., Xiamen, China), ensuring consistent lighting conditions across all visual treatments (see Fig. S3 [↗](#)).

Artificial simulation of the Earth's magnetic field

A three-dimensional pair of Helmholtz coils (Nanjing Science Sky Technology Limited) was used to manipulate the Earth's magnetic field. The coils measured 1200 × 1100 × 1000 mm, with a cold-state direct current resistance of 2 Ω per coil and an insulation resistance exceeding 10 MΩ. The magnetic field strength at the central X, Y, and Z axes exceeded 10 GS@6A (maximum current), with a spatial uniformity of 0.5% within a φ144 × 144 × 144 mm volume. At the start of each experiment, we used a TM4300B handheld three-axis fluxgate magnetometer to measure the ambient magnetic field strength and spatial vector intensity at the experimental site (averaged

over three measurements). A DH1766 series linear three channel output precision programmable DC power supply (Beijing Dahua Radio Instrument Factory, China) with adjustable current and voltage was employed to apply a controlled current to the 3D Helmholtz coils, reversing the horizontal geomagnetic component by approximately 180°. Crucially, we ensured that total magnetic field intensity (E), horizontal component intensity (H), and magnetic inclination angle (α) remained statistically unchanged between the controlled magnetic field (CMF) and the normal magnetic field (NMF) (see Fig. S1A–D). Electromagnetic noise at the experimental site was quantified by measuring the magnetic component of the time-dependent electromagnetic field across a 10 kHz resolution bandwidth. Measurements were performed using a Spectran NF-5035 spectrum analyzer (1 Hz–30 MHz, Aaronia AG) connected to an MDF-9400 magnetic-field antenna (9 kHz–400 MHz, Aaronia AG) via a UBBV-MDF960X preamplifier (9 kHz–60 MHz, 25 dB gain) and a 10-m RF cable. The frequency range from 1 to 10 MHz was measured using the external antenna-preamplifier configuration, while low-frequency noise between 2 kHz and 1 MHz was measured using the internal magnetic probe of the spectrum analyzer. During each measurement session, the spectrum analyzer continuously scanned the frequency range for 40 minutes, and values were logged using the instrument's peak-hold mode. Pronounced noise peaks were detected in the 2 kHz–1 MHz range, whereas noise levels from 1 to 10 MHz were low and relatively stable, ranging between 0.01 and 0.001 nT with only minor fluctuations (see Fig. S1E).

Experimental procedure

The orientation experiment followed the protocols established in our previous study (Chen et al., 2023). Experiments began after complete darkness in the night sky: the field population were tested after adapting to the natural dark environment for at least 60 minutes, while the lab-raised population started after adapting to the dark environment for at least 60 minutes following the light-to-dark cycle switch in the incubator, in a laboratory away from large electrical equipment, artificial noise, plants, and other potential sources of interference. The computer screen was positioned away from and facing away from the experimental setup. Weak red light was used only when attaching the tethering stalks and fixing the moth to the simulator. After the moth was fixed, a waiting period of 20 to 30 seconds was allowed for the subject to stabilize before data recording began. During this process, it was verified whether the insect could switch between clockwise and counterclockwise rotations. If the insect rotated in only one direction and could not fly stably, no further experiment was conducted. The encoder's horizontal placement was also checked to ensure it was not affected by tilting. The moth's wing vibrations were confirmed to be strong, with equal amplitude on both wings (indicating that the contact glue did not interfere with the wings). During the experiment, the insect's flight state was monitored by observing fluctuations in the pointer values on the experimental software. If the values did not change within 10 seconds, the experimenter needed to approach the flight arena and listen for any sounds of wing vibrations. If the insect stopped flying, the experimenter gently tapped the wall to stimulate the insect to continue flying. We analyzed individuals that were able to maintain fewer than four instances of wingbeats cessation during the experiment.

Statistical analysis

All statistical analyses and graph generation were performed using R (version 4.1.3), accessible at <https://www.r-project.org/>. A custom R script, incorporating bootstrap confidence intervals, was used for Moore's Modified Rayleigh Test (MMRT)—a non-parametric test increasingly applied in orientation behavior studies (Massy et al., 2021). MMRT accounts for both mean directions and directedness (vector lengths) of individuals, making it an alternative to the Rayleigh test. MMRT assesses whether a group exhibits significant directional tendencies, using the parameter R^* (a scaled alternative to r in circular regression), which quantifies the strength of orientation within a group, and the mean vector (MV), which represents the rank-weighted mean direction of the group. Polar plots display individual vectors, derived from azimuth data collected during the experiment by the encoder at a sampling rate of five times per second, and subsequently analyzed using the Rayleigh test. Each vector representing an individual moth's mean flight direction and directedness (r -value). The vector length reflects the proportion of time a moth maintains a

particular direction, ranging from $r = 0$ (completely un-oriented) to $r = 1$ (fully oriented). MMRT requires the analysis of each individual's average direction and r -value to determine the group-level mean MV. It is important to note that the magnitude of R^* is dependent on sample size.

To evaluate the directional orientation of moths in different treatment groups, we performed Rayleigh tests on individual flight directions. Subsequently, we used the Friedman test to examine whether different experimental phases within each experiment influenced the r -values of moth orientation, as this test is appropriate for comparing repeated measurements from the same individuals (Table S2 [↗](#)). In the initial test, only the Exp. Spring experiment showed significant differences. Therefore, we further compared the flight stability and r -values among its different experimental phases (Table S3 [↗](#)). Because r value exhibited non-normal distributions, the Wilcoxon signed-rank test was employed. Since these comparisons involved repeated measurements from the same individuals, pairwise comparisons were conducted. Because multiple comparisons were performed, failing to apply a correction would increase the likelihood of false positives; therefore, the Benjamini–Hochberg procedure was used to control the false discovery rate (FDR).

The Rayleigh r -values for each individual were also used to compare orientation strength across experiment. We accounted for the non-independence between Exp Dark I and Exp Dark II by using the paired Wilcoxon signed-rank test for their pairwise comparison. All other comparisons among independent experiments were performed using the Wilcoxon rank-sum test. The multiple comparisons corrected using the Benjamini–Hochberg procedure to control the false discovery rate (FDR). Detailed statistical results are provided in Table S4 [↗](#).

To evaluate flight stability, we measured the angular difference between consecutive seconds of recorded flight behavior and calculated the average directional change per second. This metric reflects the consistency and stability of individual flight orientation. Differences among treatment groups were assessed using the Wilcoxon signed-rank test (Table S5 [↗](#)) because some of the data did not follow a normal distribution. We also accounted for the non-independence between the Exp Dark I and Exp Dark II experiments, for which paired comparisons were conducted. Multiple comparisons were corrected using the Benjamini–Hochberg procedure to control the false discovery rate (FDR).

Supplementary information

Experiment	period	MV (95% CI lower-upper)	R*	n	p
Fall Field	Exp. I	183.6 (211-162)	1.76	29	$p < 0.001$
	II	191.0 (235-153)	1.16	29	$0.01 < p < 0.025$
	III	158.1 (316.3-21.2)	0.34	29	$0.5 < p < 0.9$
	IV	354.3 (331.4-27.2)	1.49	29	$0.001 < p < 0.005$
	V	167.7 (191-146)	1.62	29	$p < 0.001$
Spring Field	Exp. I	347.4 (319.5-16.7)	1.59	28	$p < 0.001$
	II	319.1 (351-283)	1.39	28	$0.001 < p < 0.005$
	III	293.7 (33.6-201.3)	0.67	28	$0.1 < p < 0.5$
	IV	245.2 (309.4-193.2)	0.95	28	$0.05 < p < 0.1$
	V	343.8 (326.56-2.92)	1.84	28	$p < 0.001$
Fall Lab	Exp. I	182.9 (203-156)	1.77	27	$p < 0.001$
	II	184.9 (222-153)	1.31	27	$0.005 < p < 0.01$
	III	111.3 (203.9-36.6)	0.69	27	$0.1 < p < 0.5$
	IV	4.1 (350.7-19.3)	2.17	27	$p < 0.001$
	V	137.6 (189.9-94.9)	1.04	27	$0.025 < p < 0.05$
Fall Control	I	180.3 (196-164)	2.10	26	$p < 0.001$
	II	184.1 (199-167)	1.94	26	$p < 0.001$
	III	195.2 (219-160)	1.43	26	$0.001 < p < 0.005$
	IV	6.1 (28.8-340.4)	1.63	26	$p < 0.001$
	V	183.0 (205-162)	1.81	26	$p < 0.001$
Fall Dark	Exp. Control	218.48 (242-187)	1.50	31	$0.001 < p < 0.005$
	Dark	41.03 (178.51-5.31)	0.55	31	$0.1 < p < 0.5$
Exp. BMVC	I	114.51 (277-18.8)	0.32	30	$0.5 < p < 0.9$
	II	215.76 (284-119)	0.65	30	$0.1 < p < 0.5$
	III	105.72 (264.8-13.9)	0.31	30	$0.5 < p < 0.9$
	IV	191.21(284-119)	0.66	30	$0.1 < p < 0.5$

Table S1. Detailed data of flight orientation behavior assay.

Table S2. Friedman test results of Rayleigh's r values across experimental

Experiment	Friedman chi-squared	df	p -value
Spring Exp. Field	16.207	4	0.002753
Fall Exp. Field	1.1714	4	0.8828
Fall Exp. Lab	8.2154	4	0.084
Fall Control	1.952	4	0.7446

Table S3. Pairwise comparison results of the mean Rayleigh's r values of Exp. Spring across different experimental phases, using the Wilcoxon signed rank test with multiple comparison correction performed by the Benjamini-Hochberg method.

Experiment	Comparison group	P value	W value
Spring Exp. Field	IV – V	0.1545	140
	III-V	0.1545	141
	II-V	0.0519	115
	I-V	0.0021	44
	III-IV	0.9153	184
	II-IV	0.5166	154
	I-IV	0.1545	84
	II-III	0.9153	210
	I-III	0.3691	122
	I-II	0.2615	116

Table S4. Multiple comparisons of mean Rayleigh test r values across experimental conditions using the Wilcoxon rank-sum test (adjusted by the Benjamini-Hochberg method).

Comparison	P value	W value
A vs B	0.32	9084
A vs C	0.01	7632
A vs D	0.54	2407
A vs E	<0.001	3252
A vs F	<0.001	17135
B vs C	0.22	7934
B vs D	0.26	2386
B vs E	<0.001	2958
B vs F	<0.001	15654
C vs D	0.0356	2543
C vs E	<0.001	3186
C vs F	<0.001	16450
D vs E	0.01	400
D vs F	<0.001	3477
E vs F	0.0700	2843

Note: A: Fall Exp. Field; B: Fall Exp. Lab; C: Fall Control; D: Fall Exp. Dark I; E: Fall Exp. Dark II; F: Fall Exp. BMVC.

Comparison group	<i>P</i> value	W value
A vs B	0.75	10006
A vs C	0.41	9967
A vs D	0.94	2227
A vs E	0.00	1479
A vs F	0.00	4237
B vs C	0.77	8957
B vs D	0.72	2005
B vs E	0.00	1376
B vs F	0.00	3890
C vs D	0.48	1850
C vs E	0.00	1119
C vs F	0.00	3199
D vs E	0.01	119
D vs F	0.00	929
E vs F	0.02	1332

Note: A: Fall Exp. Field; B: Fall Exp. Lab; C: Fall Control; D: Fall Exp. Dark-I; E: Fall Exp. Dark-II; F: Fall Exp. BMVC.

Table S5. Wilcoxon rank-sum test for mean angular velocity (directional change per second) across experimental conditions, adjusted by the Benjamini-Hochberg method.

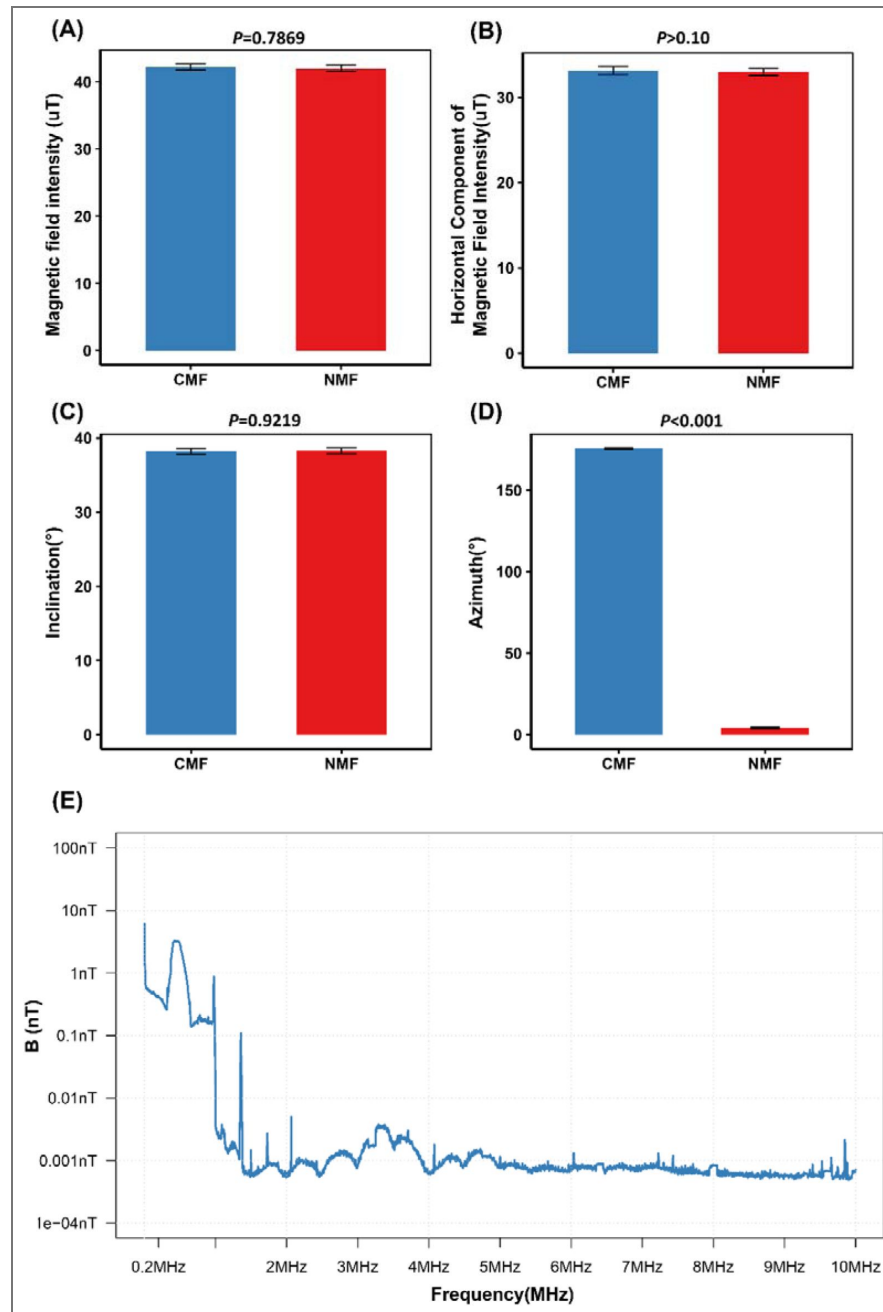


Fig. S1. The magnetic field conditions during experimental procedures in 2023 and 2024.

A Comparison of total magnetic field strength between the controlled magnetic field (CMF) and normal magnetic field (NMF) (t -test, $E_{CMF} = 42.21 \mu T$, $E_{NMF} = 42.02 \mu T$, $P = 0.7869$), showing no significant difference. **B** There is no significant difference in the intensity of the horizontal component between CMF and NMF (t -test, $H_{CMF} = 31.18$, $H_{NMF} = 31.35$, $P > 0.1$). **C** There is no significant difference in magnetic inclination between CMF and NMF ($Watson-U^2$, $\alpha_{CMF} = 38.22^\circ$, $\alpha_{NMF} = 38.28^\circ$, $P = 0.92$). **D** Comparison of magnetic azimuth between CMF and NMF, demonstrating a highly significant difference ($Watson-U^2$, $\beta_{CMF} = 175.62^\circ$, $\beta_{NMF} = 4.11^\circ$, $P < 0.001$). **E** The magnetic component B , measured in nanoteslas (nT), of the time-dependent electromagnetic field across a 10 kHz resolution bandwidth (RBW), was analyzed as a function of frequency f , expressed in megahertz (MHz). The data shown in the figure represents the maximum locked values observed over a 40-minute period at the experimental location (23.604°N, 101.977°E). Measurements were conducted using the Spectran NF5035 spectrum analyzer (1 Hz-30 MHz, Aaronia AG), MDF9400 magnetic field antenna (9 kHz-400 MHz, Aaronia AG), UBBV-MDF960X preamplifier (9 kHz-60 MHz, with a gain of 25 dB), and a 10-meter RF cable. The spectrum in the frequency range of 1 MHz to 10 MHz was first measured using the antenna paired with the preamplifier. Subsequently, low-frequency noise within the range of 2 kHz to 1 MHz was detected using the internal probe of the spectrum analyzer.

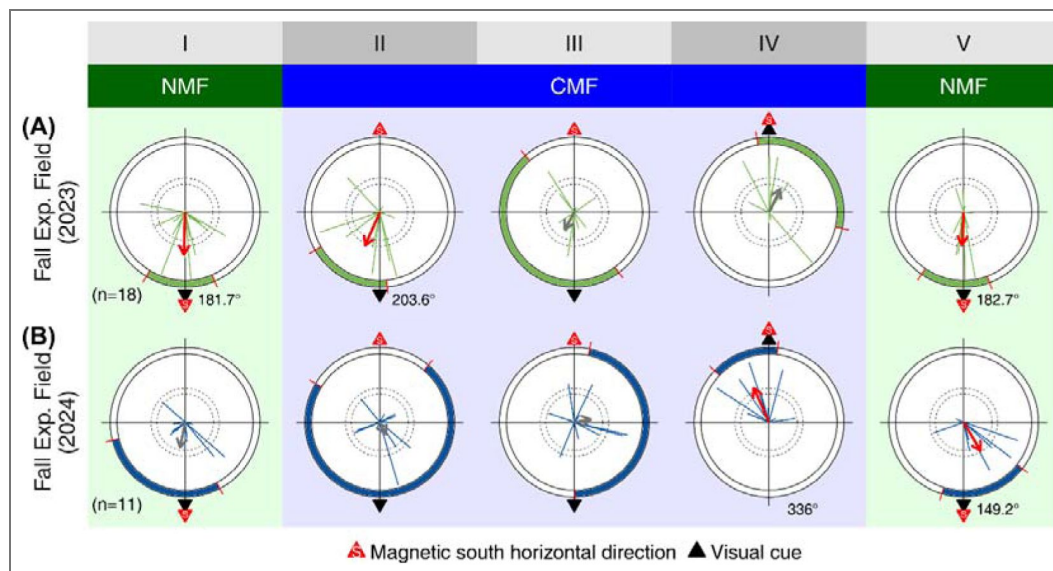


Fig. S2. Year-specific analysis of orientation behavior in field-captured armyworms during the fall migration season (2023 and 2024).

A Orientation behavior of the field-captured population in fall 2023. **B** Orientation behavior of the field-captured population in fall 2024. Experiments were conducted using fall armyworms captured from the field, revealing consistent orientation behavior across both years. In both 2023 and 2024, moths oriented significantly toward the visual cue when visual and geomagnetic cues were aligned. However, after reversing the magnetic field, the strength of directional orientation gradually diminished. Due to limited sample sizes in each year, data from both years were pooled for a comprehensive analysis, as presented in Fig. 2B.

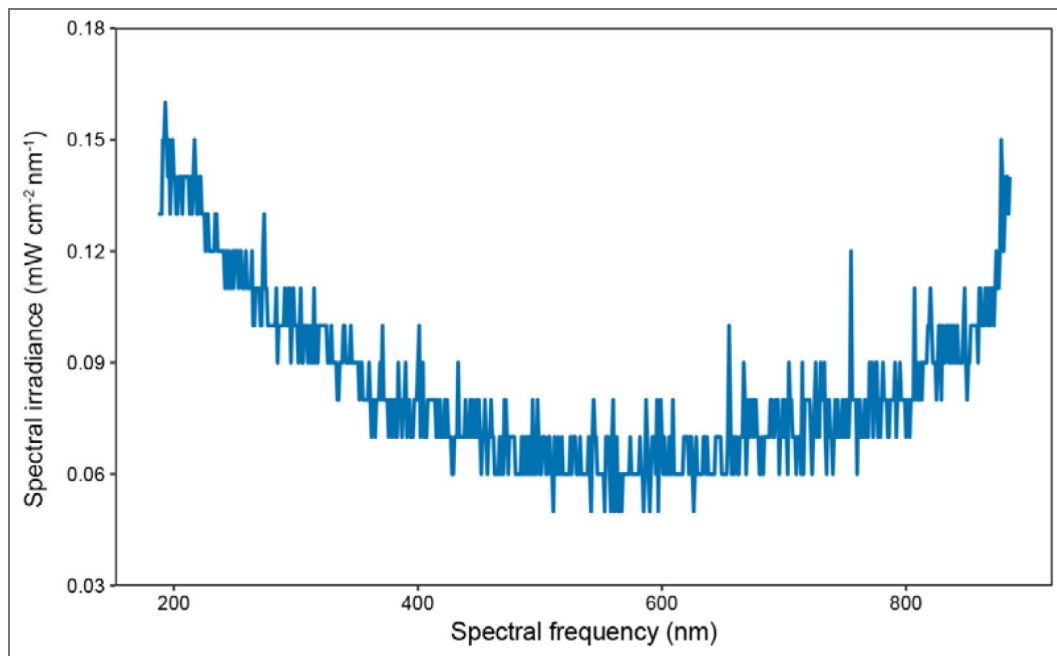


Fig. S3. Spectral distribution of light provided by the full-spectrum lamp.

The spectral distribution of the full-spectrum lamp used in the experiment was measured using an ATP2000P spectrometer (Optosky Technology Co. Ltd., Xiamen, China). The sensor was positioned vertically upward at the location originally occupied by the moth's head to measure the spectral irradiance of the light after it passed through the softbox, diffusion paper, and transparent acrylic cover. The results indicate that light radiation was detectable across most wavelengths between 200 nm and 850 nm, confirming that our experimental setup provided a full-spectrum light environment from ultraviolet to near-infrared wavelengths. This spectral range encompasses the known visual sensitivity of moths (Warrant & Somanathan, 2022). Consequently, the acrylic plate and diffusion paper did not significantly block light radiation at any specific wavelength within this range. The ambient light level in the experimental environment was measured to be below 1 lux using a Testo 540 lux meter, manufactured by Testo SE & Co. KGaA (Titisee-Neustadt, Germany).

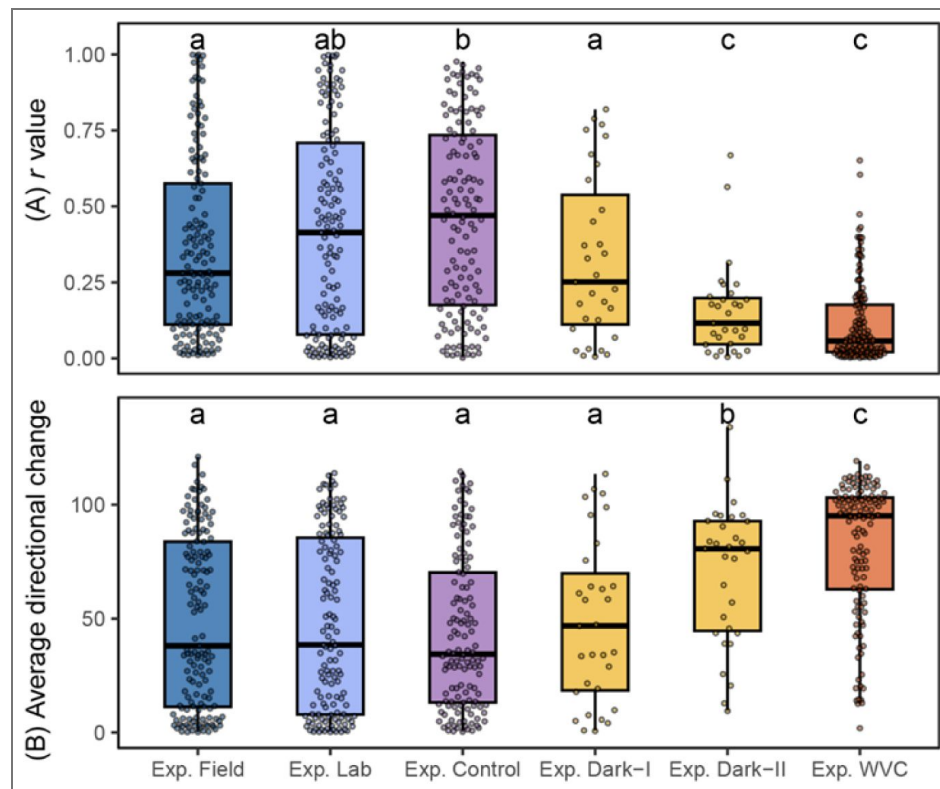


Fig. S4. The lack of visual cues and other necessary visual information significantly affects moth orientation ability and flight stability.

A Box plots show the distribution of Rayleigh test r values for individual orientations of moths in different treatment groups. **B** Box plots show the distribution of average directional degree change per second, reflecting flight stability, across treatment groups. Boxes represent the interquartile range (IQR), with the horizontal line inside indicating the median. The “whiskers” extend to the most extreme data points within 1.5 times the IQR. Error bars indicate data variability, facilitating comparison between groups. Pairwise comparisons in **A** were conducted using the Wilcoxon rank-sum test with multiple comparisons corrected by the Benjamini-Hochberg method. Detailed statistical results are provided in [Supplementary Table S4](#). In **B**, comparisons were performed using Wilcoxon rank-sum test. Groups labeled with different letters differ significantly ($P < 0.05$). Detailed statistical results are provided in [Supplementary Table S5](#).

Data availability

All data are available in the main text or the supplementary materials.

Acknowledgements

This work was supported by the National Key Research and Development Program of China (2021YFD1400700) and the Fundamental Research Funds for the Central Universities (KJJQ2025013) to G.H., and the National Natural Science Foundation of China to (32202289) B.Y.G.

Additional information

Funding

Funder	Grant reference number	Author
MOST National Key Research and Development Program of China (NKPs)	2021YFD1400700	Gao Hu
Fundamental Research Funds for Central Universities	KJJQ2025013	Gao Hu

Author ORCID iDs

Gui-Jun Wan: <https://orcid.org/0000-0002-7011-1318>

Eric J Warrant: <https://orcid.org/0000-0001-7480-7016>

Gao Hu: <https://orcid.org/0000-0002-1000-5687>

References

- Alerstam T**, Chapman JW, Bäckman J, Smith AD, Karlsson H, Nilsson C, Reynolds DR, Klaassen RHG, Hill JK (2011) Convergent patterns of long-distance nocturnal migration in noctuid moths and passerine birds. *Proc. R. Soc. B* **278**:3074-3080 <https://doi.org/10.1098/rspb.2011.0058>
- Bauer S**, Hoyer BJ (2014) Migratory animals couple biodiversity and ecosystem functioning worldwide. *Science* **344**:54-62 <https://doi.org/10.1126/science.1242552>
- Chapman JW**, Nesbit RL, Burgin LE, Reynolds DR, Smith AD, Middleton DR, Hill JK (2010) Flight orientation behaviors promote optimal migration trajectories in high-flying insects. *Science* **327**:682-685 <https://doi.org/10.1126/science.1182990>
- Chapman JW**, Reynolds DR, Wilson K (2015) Long-range seasonal migration in insects: mechanisms, evolutionary drivers and ecological consequences. *Ecol. Lett* **18**:287-302 <https://doi.org/10.1111/ele.12407>
- Chapman JW**, Nilsson C, Lim KS, Bäckman J, Reynolds DR, Alerstam T (2016) Adaptive strategies in nocturnally migrating insects and songbirds: contrasting responses to wind. *J. Anim. Ecol* **85**:115-124 <https://doi.org/10.1111/1365-2656.12420>
- Chen H**, Wang Y, Huang L, Xu CF, Li JH, Wang FY, Cheng W, Gao BY, Chapman JW, Hu G. (2022) Flight capability and the low temperature threshold of a Chinese field population of the fall armyworm *Spodoptera frugiperda*. *Insects* **13**:422 <https://doi.org/10.3390/insects13050422>
- Chen H**, Wan GJ, Li JC, Ma YB, Reynolds DR, Dreyer D, Warrant EJ, Chapman JW, Hu G. (2023) Adaptive migratory orientation of an invasive pest on a new continent. *iScience* **26**:108281 <https://doi.org/10.1016/j.isci.2023.108281>
- Chen H**, Xu CF, Wang YH, Li XR, Yu DH, Chen AD, Lyu BQ, Wu YF, Wang YM, Chapman JW, *et al.* (2025) Characteristics and seasonal variation of fall armyworm migratory behavior in their year-round breeding areas in South China. *Pest Manag Sci* **81**:4051-4062 <https://doi.org/10.1002/ps.8772>

- Drake VA, Reynolds DR (2012) *Radar entomology: observing insect flight and migration* Wallingford, UK: CABI.
- Dreyer D, Frost B, Mouritsen H, Günther A, Green K, Whitehouse M, Johnsen S, Heinze S, Warrant E (2018a) The Earth's magnetic field and visual landmarks steer migratory flight behavior in the nocturnal Australian Bogong moth. *Curr. Biol* **28**:2160-2166 <https://doi.org/10.1016/j.cub.2018.05.030>
- Dreyer D, el Jundi B, Kishkinev D, Suchentrunk C, Campostrini L, Frost BJ, Zechmeister T, Warrant EJ (2018b) Evidence for a southward autumn migration of nocturnal noctuid moths in central Europe. *J. Exp. Biol* **221**:jeb179218 <https://doi.org/10.1242/jeb.179218>
- Dreyer D, Frost B, Mouritsen H, Lefèvre A, Menz M, Warrant EJ (2021) A guide for using flight simulators to study the sensory basis of long-distance migration in insects. *Front. Behav. Neurosci* **15**:678936 <https://doi.org/10.3389/fnbeh.2021.678936>
- Dreyer D, Adden A, Chen H, Frost B, Mouritsen H., Xu J. J., Green K., Whitehouse M., Chahl J., Wallace J., et al. (2025) Bogong moths use a stellar compass for long-distance navigation at night. *Nature* **643**:994-1000 <https://doi.org/10.1038/s41586-025-09135-3>
- Fleischmann PN, Grob R, Müller VL, Wehner R, Rössler W (2018) The geomagnetic field is a compass cue in *Cataglyphis* ant navigation. *Curr. Biol* **28**:1440-1444 <https://doi.org/10.1016/j.cub.2018.03.043>
- Freas CA, Spetch ML (2023) Varieties of visual navigation in insects. *Anim Cogn* **26**:319-342 <https://doi.org/10.1016/j.conb.2011.05.020>
- Foster JJ, Smolka J, Nilsson DE, Dacke M (2018) How animals follow the stars. *Proc. R. Soc. B* **285**:20172322 <https://doi.org/10.1098/rspb.2017.2322>
- Fu P, Li X, Sun Z, Chen Y, Lu Z, Tian C, Hu G, Gui F (2025) Population genetics and trajectory simulation reveals the invasion process of the fall armyworm (*Spodoptera frugiperda*) in the Eastern Hemisphere. *Evol Appl* **18**:e70086 <https://doi.org/10.1111/eva.70086>
- Gao BY, Reynolds DR, Zhai BP, Hu G, Chapman JW (2020) The 'migratory connectivity' concept, and its applicability to insect migrants. *Move Ecol* **8**:48 <https://doi.org/10.1073/pnas.2317646121>
- Gao BY, Wotton KR, Hawkes WLS, Menz MHM, Reynolds DR, Zhai BP, Hu G, Chapman JW (2020) Adaptive strategies of high-flying migratory hoverflies in response to wind currents. *Proc. R. Soc. B* **287**:20200406 <https://doi.org/10.1098/rspb.2020.0406>
- Gao BY, Reynolds DR, Hu G, Chapman JW (2024) Effects of nocturnal celestial illumination on high-flying insect migration. *Phil. Trans. Roy. Soc. B* **379**:20230115 <https://doi.org/10.1098/rstb.2023.0115>
- Gao FQ, Chen H, Kyerematen R, Hu G, Early R, Chapman JW (2025) High migratory potential of fall armyworm in West Africa despite stable temperatures and widely available year-round habitats. *Insect Sci* <https://doi.org/10.1111/1744-7917.13502>
- Goergen G, Kumar PL, Sankung SB, Togola A, Tamo M (2016) First report of outbreaks of the fall armyworm *Spodoptera frugiperda* (J E Smith) (Lepidoptera, Noctuidae), a new alien invasive pest in west and central Africa. *PLoS One* **11**:e0165632 <https://doi.org/10.1371/journal.pone.0165632>
- Goforth KM, Lohmann CMF, Gavin A, Henning R, Harvey A, Hinton TL, Lim DS, Lohmann KJ (2025) Learned magnetic map cues and two mechanisms of magnetoreception in turtles. *Nature* **638**:1015-1022 <https://doi.org/10.1038/s41586-024-08554-y>
- Grob R, Wegmann JW, Rössler W, Fleischmann PN (2024) *Cataglyphis* ants have a polarity-sensitive magnetic compass. *Curr. Biol* **34**:5833-5838 <https://doi.org/10.1016/j.cub.2024.11.012>
- Farnworth MS, Degen J, Warrant E, Montgomery SH, el Jundi B (2025) The diversity of lepidopteran spatial orientation strategies - neuronal mechanisms and emerging challenges in a changing world. *J Comp Physiol A Neuroethol Sens Neural Behav Physiol* <https://doi.org/10.1007/s00359-025-01780-3>
- Guo J, Fu X, Zhao S, Shen X, Wyckhuys KAG, Wu K (2020) Long-term shifts in abundance of (migratory) crop-feeding and beneficial insect species in northeastern Asia. *J. Pest Sci* **93**:583-594 <https://doi.org/10.1007/s10340-019-01191-9>

- Holland RA, Wikelski M, Wilcove DS (2006) How and why do insects migrate?. *Science* **313**:794-796 <https://doi.org/10.1126/science.1127272>
- Hu G, Lim KS, Horvitz N, Clark SJ, Reynolds D. R., Sapir N., Chapman J. W. (2016) Mass seasonal bioflows of high-flying insect migrants. *Science* **354**:1584-1587 <https://doi.org/10.1126/science.aah4379>
- Hu G, Feng HQ, Otuka A, Reynolds DR, Drake VA, Chapman JW (2025) The east Asian Insect flyway: geographical and climatic factors driving migration among diverse crop pests. *Annu. Rev. Entomol* **70**:1-22 <https://doi.org/10.1146/annurev-ento-012524-124018>
- Huang JR, Feng HQ, Drake VA, Reynolds DR, Gao BY, Chen FJ, Zhang GY, Zhu J, Gao YB, Zhai BP, *et al.* (2024) Massive seasonal high-altitude migrations of nocturnal insects above the agricultural plains of East China. *Proc. Nat. Acad. Sci. USA* **121**:e2317646121 <https://doi.org/10.1073/pnas.2317646121>
- Hore PJ, Mouritsen H (2016) The radical-pair mechanism of magnetoreception. *Annu. Rev. Biophys* **45**:299-344 <https://doi.org/10.1146/annurev-biophys-032116-094545>
- Kenis M, Benelli G, Biondi A, Calatayud PA, Day R, Desneux N, Harrison RD, Kriticos D, Rwomushana I, van den Berg J, *et al.* (2023) Invasiveness, biology, ecology, and management of the fall armyworm, *Spodoptera frugiperda*. *Entomol Gen* **43**:187-241 <https://doi.org/10.1127/entomologia/2022/1659>
- Li XJ, Wu MF, Ma J, Gao BY, Wu QL, Chen AD, Liu J, Jiang YY, Zhai BP, Early R, *et al.* (2020) Prediction of migratory routes of the invasive fall armyworm in eastern China using a trajectory analytical approach. *Pest Manag Sci* **76**:454-463 <https://doi.org/10.1002/ps.5530>
- Massy R, Hawkes WLS, Doyle T, Troschianko J, Menz MHM, Roberts NM, Chapman JW, Wotton KR (2021) Hoverflies use a time-compensated sun compass to orientate during autumn migration. *Proc. R. Soc. B* **288**:20211805 <https://doi.org/10.1098/rspb.2021.1805>
- Menz MHM, Scacco M, Bürki-Spycher HM, Williams HJ, Reynolds DR, Chapman JW, Wikelski M (2022) Individual tracking reveals long-distance flight-path control in a nocturnally migrating moth. *Science* **377**:764-768 <https://doi.org/10.1126/science.abn166>
- Mouritsen H, Frost B (2002) Virtual migration in tethered flying monarch butterflies reveals their orientation mechanisms. *Proc. Nat. Acad. Sci. USA* **99**:10162-10166 <https://doi.org/10.1073/pnas.152137299>
- Mouritsen H (2018) Long-distance navigation and magnetoreception in migratory animals. *Nature* **558**:50-59 <https://doi.org/10.1038/s41586-018-0176-1>
- Nagoshi RN, Meagher RL (2008) Review of fall armyworm (Lepidoptera: Noctuidae) genetic complexity and migration. *Fla Entomol* **91**:546-554 <https://doi.org/10.1653/0015-4040-91.4.546>
- Pakhomov A, Shapoval A, Shapoval N, Kishkinev D (2025) Not all butterflies are monarchs: compass systems in the red admiral (*Vanessa atalanta*), a European diurnal migrant. *bioRxiv* <https://doi.org/10.1101/2025.05.01.651646>
- Reppert SM, Guerra PA, Merlin C (2016) Neurobiology of monarch butterfly migration. *Annu. Rev. Entomol* **61**:25-42 <https://doi.org/10.1146/annurev-ento-010814-020855>
- Srygley RB, Dudley R (2008) Optimal strategies for insects migrating in the flight boundary layer: Mechanisms and consequences. *Integr Comp Biol* **48**:119-133 <https://doi.org/10.1093/icb/icn011>
- Tay WT, Meagher RL, Czapak C, Groot AT (2023) *Spodoptera frugiperda*: Ecology, evolution, and management options of an invasive species. *Annu. Rev. Entomol* **68**:299-317 <https://doi.org/10.1146/annurev-ento-120220-102548>
- Tong DD, Zhang L, Wu NN, Xie DJ, Fang GJ, Coates BS, Sappington TW, Liu YQ, Cheng YX, Xia JX, *et al.* (2022) The oriental armyworm genome yields insights into the long-distance migration of noctuid moths. *Cell Rep* **41**:111843 <https://doi.org/10.1016/j.celrep.2022.111843>
- Wan GJ, Hayden AN, Iiams SE, Merlin C (2021) Cryptochrome 1 mediates light-dependent inclination magnetosensing in monarch butterflies. *Nat. Comm* **12**:771 <https://doi.org/10.1038/s41467-021-21002-z>
- Warrant EJ, Dacke M (2011) Vision and visual navigation in nocturnal insects. *Ann. Rev. Entomol* **56**:239-254 <https://doi.org/10.1146/annurev-ento-120709-144852>

- Warrant EJ, Frost B, Green K, Mouritsen H, Dreyer D, Adden A, Brauburger K, Heinze S (2016) The Australian Bogong moth *Agrotis infusa*: A long-distance nocturnal navigator. *Front. Behav. Neurosci* **10**:77 <https://doi.org/10.3389/fnbeh.2016.00077>
- Warrant E, Somanathan H (2022) Colour vision in nocturnal insects. *Philos Trans R Soc Lond B Biol Sci* **377**:20210285 <https://doi.org/10.1098/rstb.2021.0285>
- Werber Y, Adin E, Chapman JW, Reynolds DR, Sapir N (2025) Active navigation and meteorological selectivity drive mass intercontinental insect migration patterns through the Levant. *Proc. R. Soc. B* **292**:20250587 <https://doi.org/10.1098/rspb.2025.0587>
- Westbrook J, Fleischer S, Jairam S, Meagher R, Nagoshi R (2019) Multigenerational migration of fall armyworm, a pest insect. *Ecosphere* **10**:e02919 <https://doi.org/10.1002/ecs2.2919>
- Wu QL, Jiang YY, Liu J, Hu G, Wu KM (2021) Trajectory modeling revealed a southwest-northeast migration corridor for fall armyworm *Spodoptera frugiperda* (Lepidoptera: Noctuidae) emerging from the North China Plain. *Insect Sci* **28**:649-661 <https://doi.org/10.1111/1744-7917.12852>
- Wu MF, Qi GJ, Chen H, Ma J, Liu J, Jiang YY, Lee GS, Otuka A, Hu G (2022) Overseas immigration of fall armyworm, *Spodoptera frugiperda* (Lepidoptera: Noctuidae) invading Korea and Japan in 2019. *Insect Sci* **29**:505-520 <https://doi.org/10.1111/1744-7917.12940>
- Satterfield DA, Sillett TS, Chapman JW, Altizer S, Marra PP (2020) Seasonal insect migrations: massive, influential, and overlooked. *Front. Ecol. Environ* **18**:335-344 <https://doi.org/10.1002/fee.2217>
- Sun XX, Hu CX, Jia HR, Wu QL, Shen XJ, Zhao SY, Jiang YY, Wu KM (2021) Case study on the first immigration of fall armyworm, *Spodoptera frugiperda* invading into China. *J. Integr. Agri* **20**:664-672 [https://doi.org/10.1016/s2095-3119\(19\)62839-x](https://doi.org/10.1016/s2095-3119(19)62839-x)
- Xu JJ, Pan W, Zhang YC, Li Y, Wan GJ, Chen FJ, Sword GA, Pan WD (2017) Behavioral evidence for a magnetic sense in the oriental armyworm, *Mythimna separata*. *Biol. Open* **6**:340-347 <https://doi.org/10.1242/bio.022954>

Peer reviews

Reviewer #1 (Public review):

Summary

The manuscript by Ma et al. provides robust and novel evidence that the noctuid moth *Spodoptera frugiperda* (Fall Armyworm) possesses a complex compass mechanism for seasonal migration that integrates visual horizon cues with Earth's magnetic field (likely its horizontal component). This is an important and timely study: apart from the Bogong moth, no other nocturnal Lepidoptera has yet been shown to rely on such a dual-compass system. The research therefore expands our understanding of magnetic orientation in insects with both theoretical (evolution and sensory biology) and applied (agricultural pest management, a new model of magnetoreception) significance.

The study uses state-of-the-art methods and presents convincing behavioural evidence for a multimodal compass. It also establishes the Fall Armyworm as a tractable new insect model for exploring the sensory mechanisms of magnetoreception, given the experimental challenges of working with migratory birds. Overall, the experiments are well designed, the analyses are appropriate, and the conclusions are generally well supported by the data.

Strengths

- Novelty and significance: First strong demonstration of a magnetic-visual compass in a globally relevant migratory moth species, extending previous findings from the Bogong moth and opening new research avenues in comparative magnetoreception.

- **Methodological robustness:** Use of validated and sophisticated behavioural paradigms and magnetic manipulations consistent with best practices in the field. The use of 5 min bins to study a dynamic nature of magnetic compass which is anchored to a visual cue but updated with latency of several minutes is an important finding and a new methodological aspect in insect orientation studies.
- **Clarity of experimental logic:** The cue-conflict and visual cue manipulations are conceptually sound and capable of addressing clear mechanistic questions.
- **Ecological and applied relevance:** Results have implications for understanding migration in an invasive agricultural pest with expanding global range.
- **Potential model system:** Provides a new, experimentally accessible species for dissecting the sensory and neural bases of magnetic orientation.

Weaknesses

Overall, this is a strong study, and the authors have completed an excellent major revision that has undoubtedly addressed most major and minor issues. The remaining points below are minor recommendations, and I acknowledge that differences in opinion are always possible:

(1) Structure and Presentation of Results

- I recommend reordering the visual-cue experiments to progress from simpler conditions (no cues) to more complex ones (cue-conflict). This would improve narrative logic and accessibility for non-specialist readers. The authors have chosen not to implement this suggestion, which I respect, but my recommendation stands.

(2) Ecological Interpretation

- The authors should expand their discussion on how the highly simplified, static cue setup translates to natural migratory conditions, where landmarks are dynamic, transient, or absent. Specifically, further consideration is needed on how the compass might function when landmarks shift position, become obscured, or are replaced by celestial cues. Additionally, the discussion would benefit from a more consolidated section with concrete suggestions for future experiments involving transient, multiple, or more naturalistic visual cues.

This point was addressed partially in one paragraph of the Discussion, which reads as follows:

"In nature, they are likely to encounter a range of luminance-gradient visual cues, including relatively stable celestial cues as well as transient or shifting local features encountered en route. Although such natural cues differ from our simplified laboratory stimulus, they may represent intermittently sampled visual inputs that can be optimally integrated with magnetic information, with the congruency between visual and magnetic cues likely playing a key role in maintaining a stable compass response. Whether the cues are static or changing, brief periods without them may still allow the subsequent recovery of a stable long-distance orientation strategy. Determining which types of natural visual cues support the magnetic-visual compass, and how they interact with magnetic information, including how their momentary alignment or angular relationship is integrated and how such visual cue-magnetic field interactions may require time to influence orientation, together with elucidating the genetic and ecological bases of multimodal orientation, will be important objectives for future research."

While this paragraph is informative, the wording remains lengthy, somewhat unclear, and vague. Shorter, clearer statements would improve readability and impact. For example:

- How could moths maintain direction during periods when only the magnetic field is present and visual landmarks are absent?
- Could celestial cues (e.g., stars) compensate, and what happens if these are also obscured?
- What role does saliency play when multiple visual landmarks are present simultaneously?
- How might a complex skyline without salient landmarks affect orientation?

Including simple, concise sentences that pose concrete open questions and suggest experimental designs would strengthen the discussion without creating space issues. In my view, a comprehensive discussion of how the simplified, static cue setup relates to natural migratory conditions—where landmarks are dynamic, transient, or absent—would add significant value to the paper.

(3) Methodological Details and Reproducibility

- The lack of luminance level measurements should be explicitly highlighted.
- The authors chose not to adjust figure legends by replacing "magnetic South" with "magnetic North." While I believe this would be more conventional and preferable, this is ultimately a minor stylistic issue.

(4) Conceptual Framing and Discussion

- Although the authors made a good attempt to explain the limitations of using an artificial visual cue, I believe there is room for a more explicit argument. For example, it could be stated clearly that this species is unlikely to encounter a situation in nature where a single, highly salient landmark coincides with its migratory direction. Therefore, how these findings translate to real migratory contexts remains an open question. A sentence or two making this point directly would strengthen the discussion.

(5) Technical and Open-Science Points

- Sharing the R code openly (e.g., via GitHub) should be seriously considered. The code does not need to be perfectly formatted, but making it available would be highly beneficial from an open-science perspective.

<https://doi.org/10.7554/eLife.109098.2.sa2>

Reviewer #2 (Public review):

Summary:

The work titled "Geomagnetic and visual cues guide seasonal migratory orientation in the nocturnal fall armyworm, the world's most invasive insect" provided experimental evidence on how geomagnetic and visual cues are integrated, and visual cues are indispensable for magnetic orientation in the nocturnal fall armyworm.

Strengths:

It has been demonstrated that the Australian Bogon moth could integrate global stellar cues with the geomagnetic field for long distance navigation. However, data are lacking for other insects. This study suggested that the integration of geomagnetic and visual cues may represent a conserved navigational mechanism broadly employed across migratory insects.

Weaknesses:

The visual cues used in the indoor experimental system designed by the authors may have some limitations in ecological relevance. The author may need more explanations on this experimental system.

In the revised manuscript, the authors have added explanations in the discussion section. I am fine with the revision.

<https://doi.org/10.7554/eLife.109098.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary

*The manuscript by Ma et al. provides robust and novel evidence that the noctuid moth *Spodoptera frugiperda* (Fall Armyworm) possesses a complex compass mechanism for seasonal migration that integrates visual horizon cues with Earth's magnetic field (likely its horizontal component). This is an important and timely study: apart from the Bogong moth, no other nocturnal Lepidoptera has yet been shown to rely on such a dual-compass system. The research therefore expands our understanding of magnetic orientation in insects with both theoretical (evolution and sensory biology) and applied (agricultural pest management, a new model of magnetoreception) significance.*

The study uses state-of-the-art methods and presents convincing behavioural evidence for a multimodal compass. It also establishes the Fall Armyworm as a tractable new insect model for exploring the sensory mechanisms of magnetoreception, given the experimental challenges of working with migratory birds. Overall, the experiments are well-designed, the analyses are appropriate, and the conclusions are generally well supported by the data.

Strengths

(1) Novelty and significance: First strong demonstration of a magnetic-visual compass in a globally relevant migratory moth species, extending previous findings from the Bogong moth and opening new research avenues in comparative magnetoreception.

(2) Methodological robustness: Use of validated and sophisticated behavioural paradigms and magnetic manipulations consistent with best practices in the field. The use of 5-minute bins to study the dynamic nature of the magnetic compass which is anchored to a visual cue but updated with a latency of several minutes, is an important finding and a new methodological aspect in insect orientation studies.

(3) Clarity of experimental logic: The cue-conflict and visual cue manipulations are conceptually sound and capable of addressing clear mechanistic questions.

(4) Ecological and applied relevance: Results have implications for understanding migration in an invasive agricultural pest with an expanding global range.

(5) Potential model system: Provides a new, experimentally accessible species for dissecting the sensory and neural bases of magnetic orientation.

Weaknesses

While the study is strong overall, several recommendations should be addressed to improve clarity, contextualisation, and reproducibility:

We thank Reviewer #1 for the positive and encouraging evaluation of our study. We appreciate the recognition of our work's strengths and are grateful for the constructive feedback on the remaining weaknesses, which will guide and strengthen our revisions.

Structure and presentation of results

Requires reordering the visual-cue experiments to move from simpler (no cues) to more complex (cue-conflict) conditions, improving narrative logic and accessibility for non-specialists.

Thank you for this thoughtful suggestion. While we appreciate the rationale for presenting results from simpler to more complex conditions, we kept the original sequence because it aligns with the logic of our study. Our initial aim was to determine whether fall armyworms use a magnetic compass integrated with visual cues, as shown in the Bogong moth. After establishing this phenotype, we then examined whether visual cues are required for maintaining magnetic orientation. We have also clarified in the Introduction that magnetic orientation in the Bogong moth relies on integration with visual cues, which provides readers with clearer context and improves the overall narrative flow.

Ecological interpretation

(a) The authors should discuss how their highly simplified, static cue setup translates to natural migratory conditions where landmarks are dynamic, transient or absent.

Thank you for raising this important point. We agree that natural migratory environments provide visual information that is often dynamic, transient, or intermittently absent, in contrast to the simplified and static cue used in our indoor experiments. Our intention in using a minimal, static cue was to isolate and test the fundamental presence of magnetic–visual integration in fall armyworms under fully controlled conditions. To address the reviewer's concern, we have added a brief note in the Discussion indicating that fall armyworms may encounter both static and dynamic luminance-based visual cues in nature, such as light–dark gradients created by terrain features or more stable celestial patterns. Although these natural cues differ from our simplified laboratory stimulus, they may similarly provide asymmetric visual structure that can be integrated with magnetic information. We also note that determining which natural visual cues support the magnetic–visual compass will be an important direction for future work.

(b) Further consideration is required regarding how the compass might function when landmarks shift position, are obscured, or are replaced by celestial cues. Also, more consolidated (one section) and concrete suggestions for future experiments are needed, with transient, multiple, or more naturalistic visual cues to address this.

Thank you for this constructive suggestion. We appreciate the reviewer's point that additional consideration of how the compass might function under shifting, obscured, or celestial visual cues would strengthen the manuscript. Given the limited evidence currently available for this species, we have incorporated a concise and appropriately cautious discussion addressing these possibilities.

Methodological details and reproducibility

(a) It would be better to move critical information (e.g., electromagnetic noise measurements) from the supplementary material into the main Methods.

Thank you for this helpful suggestion. In the revised manuscript, we have added the key electromagnetic noise measurements information to the main Methods section.

(b) Specifying luminance levels and spectral composition at the moth's eye is required for all visual treatments.

Thank you for this helpful comment. We have clarified in the Methods as well as the legend of Fig. S3 that both luminance levels and spectral composition were measured at the position corresponding to the moth's head.

(c) Details are needed on the sex ratio/reproductive status of tested moths, and a map of the experimental site and migratory routes (spring vs. fall) should be included.

Thanks. We have added the reproductive status of the tested moths in the Methods, specifying that all individuals used were unmated 2-day-old adults.

(d) Expanding on activity-level analyses is required, replacing "fatigue" with "reduced flight activity," and clarifying if such analyses were performed.

Thank you for this comment. In this context, the term “fatigue” referred to the possibility that moths might gradually lose motivation or attention to orient when flying for an extended period in a simplified, artificial environment with limited sensory cues. Such a decrease in orientation motivation over time could, in theory, lead to a loss of individual orientation and consequently to the observed loss of group orientation. To test this possibility, we analyzed the orientation performance of each individual moth across different phases using the Rayleigh test. The r -value was used as a measure of individual directedness (higher r -values indicate stronger orientation). Our results showed that mean r -values did not differ significantly among the experimental phases (multiple comparisons, Table S2). This indicates that 25min measurement itself was not responsible for the loss of orientation. We did not perform a quantitative activity-level analysis in this study. However, as mentioned in Methods, flight activity was continuously monitored during the experiments by observing fluctuations in the pointer values on the experimental software, which corresponded to the moth's rotational movements. If the pointer values remained unchanged for more than 10 seconds, the experimenter checked for wing vibrations by sound; if the moth had stopped flying, gentle tapping on the arena wall was used to stimulate renewed flight. Only individuals that maintained active flight throughout the experiment, with fewer than four instances of wingbeat cessation, were included in the analysis. We also mentioned that activity level analysis was not performed due to technical difficulties in the revised manuscript.

Figures and data presentation

(a) The font sizes on circular plots should be increased; compass labels (magnetic North), sample sizes, and p -values should be included.

Thank you for this helpful suggestion. Regarding the compass labels and statistical reporting, our analysis provides significance levels as ranges rather than exact p -values; therefore, we clarified in the figure legends that the two dashed circles correspond to thresholds for statistical significance $p = 0.05$ and $p = 0.01$, respectively. Sample sizes are already indicated within each panel. To avoid visual clutter caused by displaying both magnetic North and South, we show only the magnetic South direction (mS) consistently across panels, which can improve readability.

(b) More clarity is required on what "no visual cue" conditions entail, and schematics or photos should be provided.

Thank you for this comment. In our study, the “no visual cue” condition refers to the absence of the black triangular landmark inside the flight simulator. To improve clarity, we have updated the legend of Fig. 4 to explicitly state this and have referred readers to the schematic in Fig. 1, which illustrates the structure of the flight simulator. These additions clarify what the “no visual cue” condition entails without requiring additional schematics.

(c) The figure legends should be adjusted for readability and consistency (e.g., replace "magnetic South" with magnetic North, and for box plots better to use asterisks for significance, report confidence intervals).

Thank you. Regarding the choice of compass labeling, we intentionally used magnetic South (mS) rather than magnetic North (mN) because the main population tested in our experiments represents the autumn migratory generation. During autumn, fall armyworms orient southward when visual and magnetic cues are aligned. Using magnetic South in the plots therefore provides a clearer representation of cue alignment in this season and avoids potential confusion when interpreting the combined visual–magnetic information.

Conceptual framing and discussion

(a) Generalisations across species should be toned down, given the small number of systems tested by overlapping author groups.

Thank you for this valuable comment. In the revised manuscript, we have softened such statements in both abstract and maintext.

(b) It requires highlighting that, unlike some vertebrates, moths require both magnetic and visual cues for orientation.

Thank you for this helpful suggestion. We have added a sentence to the Discussion explicitly highlighting that, unlike some vertebrates capable of using magnetic information in the absence of visual cues, moths require the integration of both magnetic and visual cues for accurate orientation. This clarification emphasizes the distinct multimodal nature of compass use in migratory moths.

(c) It should be emphasised that this study addresses direction finding rather than full navigation.

Thank you for this important clarification. We have now made it explicit in the manuscript that our experiments address direction finding (i.e., orientation) rather than full navigation. This distinction is stated in both the Introduction and Discussion to clearly define the scope of the study.

(d) Future Directions should be integrated and consolidated into one coherent subsection proposing realistic next steps (e.g., more complex visual environments, temporal adaptation to cue-field relationships).

Thank you for this constructive suggestion. We agree that outlining realistic next steps is valuable. However, given the limited scope of the current data, we have only slightly expanded the existing forward-looking statements in the Discussion.

(e) The limitations should be better discussed, due to the artificiality of the visual cue earlier in the Discussion.

Thank you for this comment. We agree that the artificiality of the visual cue is an important limitation of the present study. Rather than extending speculative discussion, we have clarified this limitation in the revised Discussion and highlighted the key questions that future work must address.

Technical and open-science points

Appropriate circular statistics should be used instead of t-tests for angular data shown in the supplementary material.

Thank you for this comment. We have addressed this point (Fig. S1) in the revised supplementary material.

Details should be provided on light intensities, power supplies, and improvements to the apparatus.

Thank you. Light intensities are reported as spectral irradiance measurements in Supplementary Materials, which provide full wavelength-resolved information for the illumination used, although a separate measurement of total illuminance (lux) was not performed. We have also added the requested information on the power supplies.

The derivation of individual r-values should be clarified.

Thanks. We have clarified in the revised manuscript.

Share R code openly (e.g., GitHub).

Thanks. We are in the process of organizing the relevant R code, but have not been able to upload it to GitHub before the current revision deadline. The code is available from the corresponding author upon request.

Some highly relevant - yet missing - recent and relevant citations should be added, and some less relevant ones removed..

Thanks. We added one recent relevant reference to the revised manuscript.

Reviewer #2 (Public review):

Summary:

This work provided experimental evidence on how geomagnetic and visual cues are integrated, and visual cues are indispensable for magnetic orientation in the nocturnal fall armyworm.

Strengths:

Although it has been demonstrated previously that the Australian Bogon moth could integrate global stellar cues with the geomagnetic field for long-distance navigation, the study presented in this manuscript is still fundamentally important to the field of magnetoreception and sensory biology. It clearly shows that the integration of geomagnetic and visual cues may represent a conserved navigational mechanism broadly employed across migratory insects. I find the research very important, and the results are presented very well.

We thank Reviewer #2 for the positive and encouraging evaluation of our study. We appreciate the recognition of our work's strengths.

Weaknesses:

The authors developed an indoor experimental system to study the influence of magnetic fields and visual cues on insect orientation, which is certainly a valuable approach for this field. However, the ecological relevance of the visual cue may be limited or unclear based on the current version. The visual cues were provided "by a black isosceles triangle (10 cm high, 10 cm 513 base) made from black wallpaper and fixed to the horizon at the

bottom of the arena". It is difficult to conceive how such a stimulus (intended to represent a landmark like a mountain) could provide directional information for LONG-DISTANCE navigation in nocturnal fall armyworms, particularly given that these insects would have no prior memory of this specific landmark. It might be a good idea to make a more detailed explanation of this question.

We appreciate the constructive feedback on the weaknesses, which will guide and strengthen our revisions. To address the reviewer's concern, we have added a brief note in the Discussion indicating that fall armyworms may encounter both static and dynamic luminance-based visual cues in nature, such as light-dark gradients created by terrain features or more stable celestial patterns. Although such natural cues differ from our simplified laboratory stimulus, they may represent intermittently sampled visual inputs that can be optimally integrated with magnetic information, whether the cues are static or changing, and brief periods without them may still allow the subsequent recovery of a stable long-distance orientation strategy.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Major to Medium Suggestions

(a) Reordering of Visual Cue Tests

The manuscript currently presents cue-conflict experiments before the simpler "no visual cue" tests. For non-specialist readers, it would be more logical to start with the basic condition (no visual cues) and then move to progressively more complex ones. This provides a clearer and more logically sound narrative.

For example, the results could first demonstrate that without visual cues, the moths fail to orient (both in darkness and uniform light), and then show that introducing a single salient cue (a triangle on the horizon) restores directed behaviour. This would help readers understand the logic of the progression and should be better integrated throughout the Results and Discussion.

Thanks. We have responded this comment in Public Reviews.

(b) Translating Key Findings to Realistic Scenarios (LL 333-344 or where suitable in Discussion, and mentioning that we utilised a reductionist principle first in Intro, but clearly articulated that it is very simplified)

The main text (eg Discussion) should address how these findings translate to real-world conditions. The experimental design used a single, highly salient, and static cue, always aligned with the migratory direction. In nature, such a consistent landmark is unlikely- mountains or other features would shift position relative to the moth's trajectory as it flies.

Key questions arise which need to be addressed:

- How would the compass system adapt to changing landmark positions as the moth moves?*
- What happens when no landmarks are visible (e.g. over flat plains or cloudy nights)?*
- Would stellar or other cues take over in such cases? Your hypotheses, please.*

Addressing these points - and proposing specific future experiments (e.g. with transient or multiple visual cues)-would strengthen the ecological relevance of the findings and show a clear way forward.

Thanks for your kind comments. We now explicitly state in the Introduction that our study employs a reductionist approach using a simplified visual environment to isolate magnetic-visual interactions. As the ecological questions raised by the reviewer cannot be addressed with the current dataset, we avoid extended speculation but have added brief clarification in the Discussion and addressed these points in the Public Reviews response. We also indicate that future work will need to examine the types of visual cues that can support magnetic orientation and how such cues couple with geomagnetic information.

Technical and Methodological Points

(a) Incomplete Methods Section

Critical technical information (e.g. electromagnetic noise measurements) currently appears only in supplementary figure legends. All such details should be included in the main Methods section if the word count allows (or include a short section in the main text with reference to more details in the supplementary material).

Thanks for your kind comments. We have addressed this as suggested in the Public Reviews.

(b) Lighting Conditions

Specify luminance levels (the amount of light emitted and passing through in quanta per unit of surface, eg m2) at the moth's eye and indicate whether spectral composition was consistent between treatments (with and without the visual cue).

Thanks for your comments. We have responded to this point in the Public Reviews.

(c) Figures

- Increase font sizes on circular histograms.*
- Add compass labels (ideally magnetic North, mN, not south, etc, as it is usual in pertinent literature), sample sizes, and p-values on each panel.*
- Replace "magnetic South" (mS) indicators with magnetic North (mN) to align with convention.*

Thanks for your comments. We have responded to this point in the Public Reviews.

(d) Migratory Expectations

Include expected compass bearings for spring and autumn migrations (with citations) to relevant figures (Figure 2, 4, S2).

Thanks for your comments. We have added the information that “We recently found that fall armyworms from the year-round range in Southwest China (Yunnan) exhibit seasonally appropriate migratory headings when flown outdoors in virtual flight simulators, heading northward in the spring and southward in the fall, and this seasonal reversal is controlled by photoperiod (Chen et al., 2023).” in Introduction. Thus, we didn’t offer expected seasonal compass bearings in Results section.

(e) Add a map showing the experimental site and known migratory routes, clearly labelling spring vs fall routes. It would help justify expected headings.

Thank you for this suggestion. At present, there are no experimentally validated migratory routes (e.g., through mark-release-recapture or tracking approaches) for the specific fall armyworm population used in our study. Because these routes have not been biologically confirmed, we didn't offer a presumed migratory map that may imply unwarranted certainty.

(f) Composition of Test Groups

Indicate sex ratios and reproductive status (mated/unmated) of tested moths, if known or comment if unknown, as both can affect migratory motivation and behaviour.

Thank you for this suggestion. We have responded to this point in the Public Reviews.

(g) Role and Nature of Visual Cues

While the results clearly show that orientation disappears without visual cues, the triangle cue is highly artificial. Well-studied Bogong moths are known to rely on views of Australian mountain ranges during their nocturnal migrations, but there is no evidence that armyworms use a similar strategy. Even for bogongs, it is not just one salient mountain always in front of them on migration. Discuss whether Fall Armyworm would encounter comparable natural cues in the field along their migratory route, or whether the triangle might simply provide a frame of reference rather than a true landmark.

Thank you for this comments. We have responded to this point in the Public Reviews.

(h) Future work could test:

- More naturalistic sky cues (moonlight, star fields).

- Varying the landmark's position relative to the magnetic field - slowly moving along - transient landmarks. Also, less salient landmarks and a more complex skyline, as it is usually more complex than just a single salient peak.

Thank you for this comments. We have responded to this point in the Public Reviews. Brief discussion as suggested has been added to the revised manuscript.

Minor Comments and Line-by-Line Suggestions

L70 - Check citation (possibly Mouritsen 2018). Missing in the list of references.

Thanks. This point has been addressed.

L75 - Consider citing the new and highly relevant preprint:

Pakhomov, A., Shapoval, A., Shapoval, N., & Kishkinev, D. (2025). Not All Butterflies Are Monarchs: Compass Systems in the Red Admiral (Vanessa atalanta). bioRxiv.

Thanks. We have cited this reference.

LL81-82 - Clarify vague phrasing; specify criteria for "good" vs "poor" orientation ability. Or reword/leave out.

Thanks for your comments.

L85 - "but one," not "bar one."

Thanks. Corrected.

L124 - The 2 genetic citations are weakly linked to magnetoreception. We do not have a clear understanding of the insect magnetoreceptor and its underlying mechanism, so we simply cannot interpret genetic associations very well to underpin them to

magnetoreception. For example, does noctuid's magnetic sense require a magnetised-based receptor and genes involved in biomineralization? Consider removing or softening claims.

Thanks. Addressed.

LL123-126 - Define what for YOU constitutes "strong evidence" for magnetoreception (e.g. adaptive directional behaviour consistent with migratory orientation?). Is there such a thing as strong evidence at all?

Thanks for your comments. We agree that terms such as “confirmed” or “strong evidence” can overstate the certainty of magnetoreception findings, given the ongoing debates in the field. In the revised manuscript, we have toned down.

L153 - Indicate whether coils in NMF condition were powered or inactive.

Thanks for your comments. Addressed.

L163 - Justify use of multiple 5-min phases (e.g. temporal resolution of behaviour). It is confusing at the start, where first mentioned, and becomes clearer only towards the end, but it should be clearer at the start.

Thanks for your comments. The assay was divided into these 5-min segments to provide the temporal resolution needed to detect changes in flight orientation as the relative alignment of magnetic and visual cues was systematically altered. We now clarify this earlier in the Results.

LL167-171 - This is a good place where you can provide a map (main or supplementary with referencing) showing the study site and migration routes.

Thanks for your suggestion. We have responded to this point in the Public Reviews.

L174 - Avoid repetition of "expected."

Thanks. Addressed.

LL176-177 - Report 95% confidence intervals or equivalent and clarify which test (e.g. Moore's paired test) each p-value refers to.

Thanks for your suggestion.

LL189-191 - explain what fatigue means. I would remove fatigue and substitute it with "lowered flight activity". Also, the same statement comes later, so avoid repetitiveness and remove it in one place. The analysis of directedness is good throughout, but what about the analysis of activity level? Could you explain whether you did it or not, and if not, why, or if angular changes can serve as an activity proxy? Replace "fatigue" with "reduced flight activity." Avoid repetition. Clarify if activity level analysis was performed or if it was not, e.g. due to technical difficulties.

Thanks for your comments. We have responded to this point in the Public Reviews.

L196 - Note whether 95% CI overlaps with the expected direction. This is a crucial outcome.

Thanks for your comments.

LL203-205 - unclear, better to stick to "congruency", especially "initial congruency for the relationship between mN and visual cue" throughout.

Thanks for your suggestions.

L206 - Better to introduce a new subheading: "Laboratory-Reared Animals."

Thanks for your suggestion. A new subheading has been added in the revised manuscript.

LL207-208 - Clarify which cues were available in Chen et al. (2023) and how they differ here.

Thanks for your comments. In Chen et al. (2023), the moths oriented under an artificial starry sky together with optic flow cues. In contrast, our experiments intentionally removed both the starry-sky pattern and optic flow to avoid introducing additional visual information when testing magnetic-visual integration for orientation. We have added further clarification regarding the conditions used in Chen et al. (2023) in the revised manuscript.

L228 - Use "lab-reared" consistently throughout the entire MS. Do not mix with lab-raised.

Thanks. Addressed by consistently using "lab-raised".

Figure 2 - Confusing in parts, especially for people coming from birds and other vertebrates orientation background. At 12 o'clock, you usually expect either mN / gN (magnetic or geographic North) or the animal's own initial directional response used as control to compare the same animal's direction post-treatment. Here, your 6 o'clock is magnetic South in the first place - non-conventional. At 12 o'clock, better use mN or gN. Avoid using non-conventional references such as magnetic south. Remind readers of seasonally appropriate headings and refer to the map.

Thanks. We have responded to this point in the Public Reviews.

LL232-234 - Emphasize that cue-magnetic congruency is key. Highlight the most important point that the congruency between the seasonal migratory direction and visual cues is key, not that in spring/fall, visual cues must be towards or opposite to the migratory goal. But the visual cue could be in the migratory direction or opposite, or at an angle - this is for future direction.

Thanks. We have responded to this point in the Public Reviews.

Figure 2 and associated main text - highlight that you only tested the designs when in all seasons the salient and single visual cue was in the migratory direction (in spring it coincided with mN but in fall it was towards the magnetic south). Other directions of visual cues have not been tested, but for simplicity and consistency, you chose to do these ones as the first step, perhaps.

Thank you for this insightful comment. Yes, our experiments tested only the conditions in which the salient and single visual cue was aligned with the migratory direction. Other angular relationships between visual cues and the magnetic field were not examined in this study. For simplicity and consistency, we focused on this alignment as a first step toward understanding magnetic-visual cue integration in migratory orientation. We now highlight this in the Fig. 2 legend.

Figures captures/legends - hard to tell from the main text now, better to italicize figure caption text and visually space them from the main text.

Thanks for your suggestions.

LL 250-251 - mention to people more familiar with r - lowercase - what is the expected range for R uppercase. It is not bound 0-1 as r . Could it be negative? How large can it be?

Thanks. Thanks for the comment. After revisiting Moore (1980) we think that R^* cannot take negative values. However, since $R^* = R^*/N^{(3/2)}$, it is not bounded between 0 and 1. We didn't find any concept of an upper bound in the paper (<https://doi.org/10.2307/2335330> )

Figure 3 - Consider adding a horizontal line indicating the 5% significance threshold.

Thanks for your suggestions.

L 261 - need to have some narrative after the subheading before you insert Figure 3.

Thanks. Addressed.

LL274-275 - highlight that the timeline of this congruency between mN and a landmark and the effect of this on directedness is not explored here, but worth doing in future. How long does a new congruency or a relationship between mN and a visual cue need to be exposed to the animal to regain its directional response? Clearly, it is just a question of time of exposure so that a new association is established. Suggest future work on time-dependent adaptation to new cue-field relationships.

Thanks for your suggestion. We have now included this point as a future direction in the revised Discussion.

Figure 4 & S4 - Replace letters with asterisks/brackets for significance. The use of the letter is confusing and unconventional.

Thanks for your suggestion.

Figure 4 caption - Clarify the main takeaway.

Thanks for your suggestion.

Figure 4 - bare minimum is confusing. I understand that you wanted to avoid "no visual cues" because, as long as the animal sees things, there are things to be used as visual cues, even if this is not the intention of the experimenter. However, it needs clarification and rewording. Better to be more specific, like "no black triangle and horizon were used, just the uniformly white cylinder", or something like that.

Thanks for your comments. In our setup it accurately describes the intentional removal of both the black triangle and the horizon, leaving only the uniformly white cylinder as the visual environment. This wording was chosen to reflect the practical limitations of producing a perfectly symmetrical flight simulator under laboratory conditions, and we therefore prefer to retain the original phrasing.

L328 - Remove Xu et al. (2021) citation (not relevant). This is an in vitro study with a protein which may not work exactly as it is claimed in the paper in vivo.

Thanks. Citation removed.

L349-350 - Clarify what "no visual cue" means (e.g., uniformly white cylinder, no horizon line). Include a photo or a schematic of the inner surface of the cylinder for this condition in the Supplementary Materials.

Thanks. We have responded to this point in the Public Reviews.

L380 & throughout - Replace "barely minimum visual cues" (BMVC) with "no visual cues", clarifying limitations in Methods, meaning that you can explain that absolutely no visual

cues is practically impossible because, as long as there is light, animals can use some asymmetries as cues even if this is not the intention of the experimenter.

Thank you for this comment. We have decided to retain the term “barely minimum visual cues (BMVC)” because it accurately describes our experimental condition, which is distinct from a true “no visual cues” environment. In the revised Figure legend, we now clarify that BMVC refers to conditions in which obvious visual cues (i.e., features such as the black triangle in Fig. 1) were removed, while acknowledging that complete elimination of all visual information is not possible under illuminated conditions.

L396 - Be cautious when generalizing from two species tested by a research group that is not absolutely independent (some authors in bogong and armyworm works overlap). We saw examples in diurnal migratory butterflies (Monarchs), a more studied species than the armyworm, that the findings do not entirely translate to Red Admirals (Pakhomov et al. 2025 preprint mentioned). Suggestion to tone down any claims of broad generalisation throughout the manuscript.

Thank you for this comment. We have responded to this point in the Public Reviews.

LL402-407 - Note that, unlike birds (e.g. European robins), moths appear to require both magnetic and visual cues for orientation, whereas birds, mole rats and some other animals can use magnetic cues alone.

Thank you for this comment. We have responded to this point in the Public Reviews.

L410 - Specify that this is correct only in the Northern Hemisphere.

Thank you for this comment. Addressed.

LL415-416 - Acknowledge artificiality of single-cue setup (see the major comments above); integrate earlier in the Discussion.

Thank you for this comment. We have responded to this point in the Public Reviews.

LL420-425 - Consolidate Future Directions into a single subsection; include more concrete experimental ideas, for example, using more naturalistic, numerous transient landmarks (could be done in a virtual maze with LEDs on the wall of the cylinder with cues moving with time). Multiple visual cues. Manipulating with salience of cues - less simplistic, less salient.

Thank you for this comment. We have responded to this point in the Public Reviews.

L431 - Does this paper support this statement? I think it just tested the use of stellar cues in a zero magnetic field. It also dealt with direction finding, not navigation, which is a position-finding ability - a much more complex feat and might not be the ability of moths (requires further studies like with geographic and magnetic displacements, etc). Reword and check this. Show the distinction between direction finding and navigation.

Thank you for this comment. We have reworded the relevant sentence to use “orientation” instead of “navigation”.

L436-437 - Specify "global visual cues" (stellar, lunar, etc.) and merge all future directions into one coherent section.

Thank you for this comment. Addressed.

LL443-446 - A bit early to plan such studies because migratory direction could well be a complex multigenetic trait, so that you cannot approach it simply with the knock out of a

single gene. The genetic basis of magnetic direction needs to be first demonstrated, which leads you to the Future Directions section.

Thank you for this helpful comment. We fully agree that migratory direction is likely a complex multigenic trait, and our intention was not to imply that knocking out a single gene would be sufficient to explain magnetic or migratory orientation. Our statement aimed only to highlight that identifying candidate genes is an important first step toward understanding the genetic basis of magnetic orientation.

Line 496 - Clarify whether optic flow was used (unlike previous studies).

Thank you for pointing this out. Clarified.

LL499-511 - Clarify the improvements done in Chen's system and their relevance.

Thank you for pointing this out. We reworded this sentence "The Flash flight simulator system was developed based on the early design of the Mouritsen-Frost flight simulator and adapted for our experiments in Yuanjiang".

Line 531 - Report and compare light intensities between indoor and outdoor experiments.

Thanks for this comment. Unfortunately, due to the sensitivity limits of our current equipment, we were unable to reliably measure outdoor light intensities at night. However, we did not perform any open-top outdoor flight-simulator experiments; instead, we used field-captured moths but conducted all behavioral tests indoors.

L549 - Add make/model of power supplies.

Thanks. Addressed.

LL582-585 - Specify whether R code will be shared; recommend open access (e.g., GitHub, other open repositories). Reiterate the importance of open science and sharing all scripts. Also here, add citations to some studies where MMRT has been used recently.

Thank you for this comment. We have responded to this point in the Public Reviews.

Line 592 - Explain how individual r-values were derived from optical encoder data.

Thank you for this comment. Addressed.

L842-843 - t-tests are inappropriate for angular data; use circular tests (Watson-Williams, Mardia-Watson-Wheeler, etc.).

Thank you for this comment. Addressed.

L865 - Reword to avoid repetition of "fall." Example: "In field captured armyworms during fall migration".

Thank you for this comment. Addressed.

LL882-885 - Improve phrasing and language here. Confirming that - no colon after. "Both the acrylic plate and diffusion paper." Confirm relevance of spectra to moth visual sensitivity - add relevant citation to original studies showing that.

Thank you for this comment. Addressed.

L886 - Reword "uniform" - does not look uniform to me.

Thank you for this comment. Addressed.

Reviewer #2 (Recommendations for the authors):

The first two sentences of the abstract ("The navigational mechanisms employed by nocturnal insect migrants remain to be elucidated in most species. Nocturnal insect migrants are often considered to use the Earth's geomagnetic field for navigation, yet the underlying mechanisms of magnetoreception in insects remain elusive") are somewhat redundant. The authors may consider rewriting them.

Thank you for pointing this out. We have rewritten this opening to provide a more concise and non-repetitive introduction.

<https://doi.org/10.7554/eLife.109098.2.sa0>